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JEE MAIN · PHYSICS · CHAPTER-WISE PYQS

Thermodynamics

JEE Main Physics · Previous Year Questions with Answer Key

231	176	2002 – 2026	46	English
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MCQ (Single Correct Answer) 159 questions

Q1. JEE Main 2026 (Online) 8th April Evening Shift

One mole of diatomic gas having rotational modes only is kept in a cylinder with a piston system. The cross-section area of the cylinder is $4\ \mathrm{cm}^2$. The gas is heated slowly to raise the temperature by $1.2^{\circ}\mathrm{C}$ during which the piston moves by 25 mm . The amount of heat supplied to the gas is ____ J.

(Atmospheric pressure $= 100\mathrm{kPa}, R = 8.3\ \mathrm{J/mol.\ K}$) (Neglect mass of the piston)

(A) 24.8 (B) 10.96 (C) 15.04 (D) 29.98

Open solution

Q2. JEE Main 2026 (Online) 8th April Evening Shift

Initial pressure and volume of a monoatomic ideal gas are P and V. The change in internal energy of this gas in adiabatic expansion to volume $V_{\text{final}} = 27\ \mathrm{V}$ is ____ J.

(A) $-2PV(3\sqrt{3}-1)$ (B) $\frac{4}{3}PV$ (C) $-\frac{4}{3}PV$ (D) $\frac{3}{4}PV$

Open solution

Q3. JEE Main 2026 (Online) 6th April Evening Shift

A cylinder with adiabatic walls is closed at both ends and is divided into two compartments by a frictionless adiabatic piston. Ideal gas is filled in both (left and right) the compartments at same P, V, T. Heating is started from left side until pressure changes to $27\mathrm{P}/8$. If initial volume of each compartment was 9 litres then the final volume in right-hand side compartment is ____ litres. (for this ideal gas $\mathrm{C_P}/\mathrm{C_V} = 1.5$)

(A) 3 (B) 4 (C) 14 (D) 9

Open solution

Q4. JEE Main 2026 (Online) 5th April Evening Shift

An ideal gas at pressure P and temperature T is expanding such that $PT^3 =$ constant. The coefficient of volume expansion of the gas is ____ .

(A) $\frac{2}{T}$ (B) $\frac{1}{T}$ (C) $\frac{4}{T}$ (D) $\frac{3}{T}$

Open solution

Q5. JEE Main 2026 (Online) 5th April Morning Shift

Consider the following statements:

A. Zeroth law of thermodynamics gives concept of temperature

B. First law of thermodynamics gives concept of internal energy

C. In isothermal expansion of ideal gas, $\Delta Q \neq \Delta W$

D. Product of intensive and extensive variables is extensive

E. The ratio of any extensive variable to mass will be an extensive variable

Choose the correct combination of statements from the options given below:

(A) C, D and E Only (B) A, B and C Only (C) A, B and D Only (D) B, C and D Only

Open solution

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Q6. JEE Main 2026 (Online) 4th April Morning Shift

Open solution

An ideal gas undergoes a process maintaining relation between pressure (P) and $\text{volume}(V)$ as $P = P_o\left(1 + \left(\frac{V_o}{V}\right)^2\right)^{-1}$, where P_o and V_o are constants. If two samples A and B (two moles each) with initial volumes V_o and $3V_o$ respectively undergo above mentioned process and attain same pressure, then the difference at the temperatures of these samples, $T_B - T_A$ is ____ .

($R =$ gas constant)

(A) $\frac{9P_oV_o}{8R}$

(B) $\frac{11P_oV_o}{10R}$

(C) $\frac{7P_oV_o}{6R}$

(D) $\frac{13P_oV_o}{11R}$

Q7. JEE Main 2026 (Online) 2nd April Morning Shift

Open solution

Heat is supplied to a diatomic gas at constant pressure. Then the ratio of $\Delta Q : \Delta U : \Delta W$ is _____.

(A) 2 : 3 : 5

(B) 5 : 3 : 2

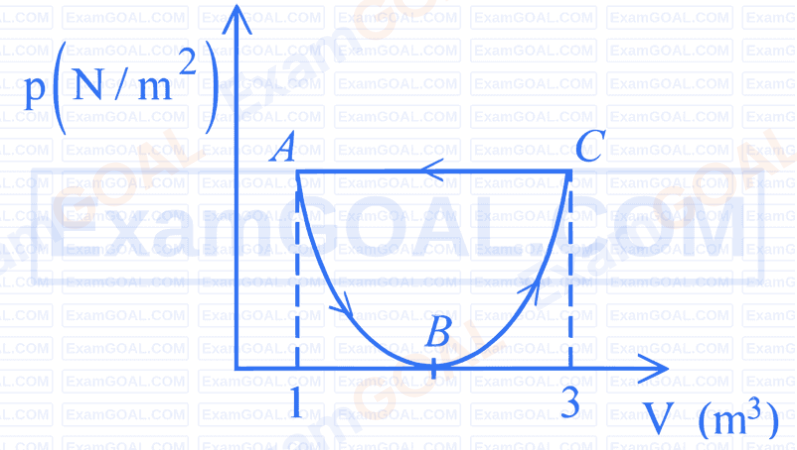
(C) 2 : 5 : 7

(D) 7 : 5 : 2

Q8. JEE Main 2026 (Online) 28th January Morning Shift

Open solution

In the following $p - V$ diagram the equation of state along the curved path is given by $(V-2)^2 = 4ap$ where a is a constant. The total work done in the closed path is



(A) $+\frac{1}{3a}$

(B) $-\frac{1}{a}$

(C) $\frac{1}{2a}$

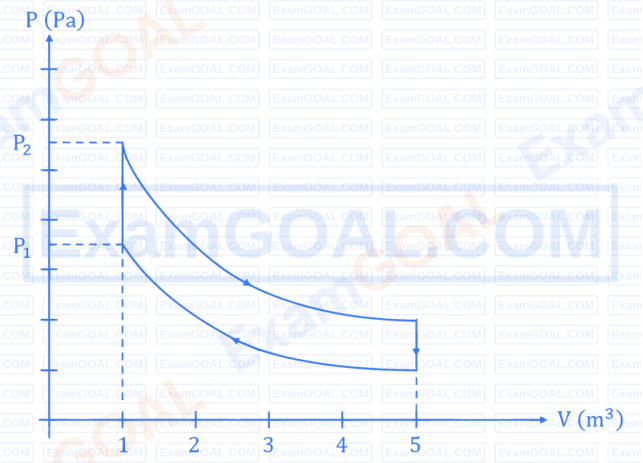
(D) $-\frac{1}{3a}$

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Q9. JEE Main 2026 (Online) 24th January Evening Shift

Open solution

10 mole of an ideal gas is undergoing the process shown in the figure. The heat involved in the process from P_1 to P_2 is α Joule ($P_1 = 21.7\ \text{Pa}$ and $P_2 = 30\ \text{Pa}, \text{C}_v = 21\ \text{J/K.mol}, R = 8.3\ \text{J/mol.K}$). The value of α is ____ .



(A) 21 (B) 28 (C) 24 (D) 15

Q10. JEE Main 2026 (Online) 24th January Morning Shift

Open solution

Density of water at 4°C and 20°C are $1000\ \text{kg/m}^3$ and $998\ \text{kg/m}^3$ respectively. The increase in internal energy of 4 kg of water when it is heated from 4°C to 20°C is ____ J.

(specific heat capacity of water $= 4.2\ \text{kJ/kg K}$ and 1 atmospheric pressure $= 10^5\ \text{Pa}$)

(A) 268799.2 (B) 315826.2 (C) 234699.2 (D) 258700.8

Q11. JEE Main 2026 (Online) 23rd January Evening Shift

Open solution

One mole of an ideal diatomic gas expands from volume V to $2V$ isothermally at a temperature 27°C and does W joule of work. If the gas undergoes same magnitude of expansion adiabatically from 27°C doing the same amount of work W, then its final temperature will be (close to) ____ $^\circ\text{C}$.

$(\log_e 2 = 0.693)$

(A) -56 (B) -117 (C) -30 (D) -189

Q12. JEE Main 2026 (Online) 23rd January Evening Shift

Open solution

The internal energy of a monoatomic gas is 3nRT. One mole of helium is kept in a cylinder having internal cross section area of $17\ \text{cm}^2$ and fitted with a light movable frictionless piston. The gas is heated slowly by suppling 126 J heat. If the temperature rises by 4°C, then the piston will move ____ cm.

(atmospheric pressure $= 10^5\ \text{Pa}$)

(A) 1.55 (B) 14.5 (C) 15.5 (D) 1.45

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Q13. JEE Main 2026 (Online) 22nd January Morning Shift

The volume of an ideal gas increases 8 times and temperature becomes $(1/4)^{th}$ of initial temperature during a reversible change. If there is no exchange of heat in this process $(\Delta Q = 0)$ then identify the gas from the following options (Assuming the gases given in the options are ideal gases) :

Open solution

(A) NH_3 (B) O_2 (C) CO_2 (D) He

Q14. JEE Main 2025 (Online) 8th April Evening Shift

A monoatomic gas having $\gamma = \frac{5}{3}$ is stored in a thermally insulated container and the gas is suddenly compressed to $\left(\frac{1}{8}\right)^{th}$ of its initial volume. The ratio of final pressure and initial pressure is:

(γ is the ratio of specific heats of the gas at constant pressure and at constant volume)

Open solution

(A) 16 (B) 32 (C) 28 (D) 40

Q15. JEE Main 2025 (Online) 7th April Evening Shift

Match **List - I** with **List - II**.

Open solution

List - I	List - II
(A) Isothermal	(I) ΔW (work done) = 0
(B) Adiabatic	(II) ΔQ (supplied heat) = 0
(C) Isobaric	(III) ΔU (change in internal energy) ≠ 0
(D) Isochoric	(IV) ΔU = 0

Choose the **correct** answer from the options given below :

(A) (A)-(III), (B)-(II), (C)-(I), (D)-(IV)

(B) (A)-(II), (B)-(IV), (C)-(I), (D)-(III)

(C) (A)-(IV), (B)-(II), (C)-(III), (D)-(I)

(D) (A)-(IV), (B)-(I), (C)-(III), (D)-(II)

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Q16. JEE Main 2025 (Online) 4th April Evening Shift

Open solution

Match List - I with List - II.

	List - I		List - II
(A)	Isobaric	(I)	$\Delta Q = \Delta W$
(B)	Isochoric	(II)	$\Delta Q = \Delta U$
(C)	Adiabatic	(III)	$\Delta Q =$ zero
(D)	Isothermal	(IV)	$\Delta Q = \Delta U + P\Delta V$

$\Delta Q =$ Heat supplied

$\Delta W =$ Work done by the system

$\Delta \mathrm{U} =$ Change in internal energy

$\mathrm{P} =$ Pressure of the system

$\Delta \mathrm{V} =$ Change in volume of the system

Choose the correct answer from the options given below :

(A) (A)-(IV), (B)-(I), (C)-(III), (D)-(II)

(B) (A)-(IV), (B)-(II), (C)-(III), (D)-(I)

(C) (A)-(IV), (B)-(III), (C)-(II), (D)-(I)

(D) (A)-(II), (B)-(IV), (C)-(III), (D)-(I)

Q17. JEE Main 2025 (Online) 3rd April Evening Shift

Open solution

An ideal gas exists in a state with pressure P_0, volume V_0. It is isothermally expanded to 4 times of its initial volume (V_0), then isobarically compressed to its original volume. Finally the system is heated isochorically to bring it to its initial state. The amount of heat exchanged in this process is

(A) $\mathrm{P}_0\,\mathrm{V}_0(\ln 2 - 0.75)$

(B) $\mathrm{P}_0\,\mathrm{V}_0(2\ln 2 - 0.75)$

(C) $\mathrm{P}_0\,\mathrm{V}_0(2\ln 2 - 0.25)$

(D) $\mathrm{P}_0\,\mathrm{V}_0(\ln 2 - 0.25)$

Q18. JEE Main 2025 (Online) 3rd April Morning Shift

Open solution

A gas is kept in a container having walls which are thermally non-conducting. Initially the gas has a volume of $800\ \mathrm{cm}^3$ and temperature $27^\circ\mathrm{C}$. The change in temperature when the gas is adiabatically compressed to $200\ \mathrm{cm}^3$ is:

(Take $\gamma = 1.5$; γ is the ratio of specific heats at constant pressure and at constant volume)

(A) 300 K **(B)** 600 K **(C)** 327 K **(D)** 522 K

Q19. JEE Main 2025 (Online) 3rd April Morning Shift

Open solution

During the melting of a slab of ice at 273 K at atmospheric pressure :

(A) Internal energy of ice-water system remains unchanged.

(B) Positive work is done by the ice-water system on the atmosphere.

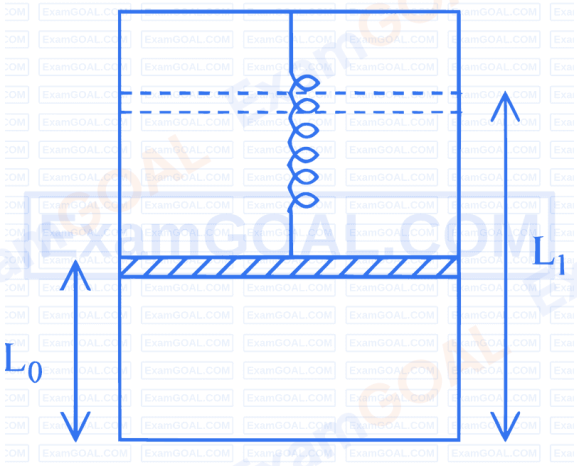
(C) Positive work is done on the ice-water system by the atmosphere.

(D) Internal energy of the ice-water system decreases.

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Q20. JEE Main 2025 (Online) 3rd April Morning Shift

Open solution



A piston of mass M is hung from a massless spring whose restoring force law goes as $F = -kx^3$, where k is the spring constant of appropriate dimension. The piston separates the vertical chamber into two parts, where the bottom part is filled with ' n ' moles of an ideal gas. An external work is done on the gas isothermally (at a constant temperature T) with the help of a heating filament (with negligible volume) mounted in lower part of the chamber, so that the piston goes up from a height L_0 to L_1, the total energy delivered by the filament is:(Assume spring to be in its natural length before heating)

(A) $nRT\ln\left(\frac{L_1}{L_0}\right) + Mg\left(L_1 - L_0\right) + \frac{3k}{4}\left({L_1}^4 - {L_0}^4\right)$

(B) $nRT\ln\left(\frac{L_1}{L_0}\right) + Mg\left(L_1 - L_0\right) + \frac{k}{4}\left(L_1^4 - {L_0}^4\right)$

(C) $nRT\ln\left(\frac{L_1^2}{L_0^2}\right) + \frac{Mg}{2}\left(L_1 - L_0\right) + \frac{k}{4}\left(L_1^4 - {L_0}^4\right)$

(D) $3nRT\ln\left(\frac{L_1}{L_0}\right) + 2Mg\left(L_1 - L_0\right) + \frac{k}{3}\left({L_1}^3 - {L_0}^3\right)$

Q21. JEE Main 2025 (Online) 2nd April Evening Shift

Open solution

Identify the characteristics of an adiabatic process in a monoatomic gas.
(A) Internal energy is constant.
(B) Work done in the process is equal to the change in internal energy.
(C) The product of temperature and volume is a constant.
(D) The product of pressure and volume is a constant.
(E) The work done to change the temperature from T_1 to T_2 is proportional to $(\mathrm{T}_2 - \mathrm{T}_1)$.
Choose the correct answer from the options given below :

(A) (B), (D) only

(B) (B), (E) only

(C) (A), (C), (E) only

(D) (A), (C), (D) only

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Q22. JEE Main 2025 (Online) 2nd April Morning Shift

Open solution

In an adiabatic process, which of the following statements is true?

(A) The internal energy of the gas decreases as the temperature increases

(B) The molar heat capacity is zero

(C) Work done by the gas equals the increase in internal energy

(D) The molar heat capacity is infinite

Q23. JEE Main 2025 (Online) 29th January Evening Shift

Open solution

Given below are two statements. One is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A) : With the increase in the pressure of an ideal gas, the volume falls off more rapidly in an isothermal process in comparison to the adiabatic process.

Reason (R) : In isothermal process, PV = constant, while in adiabatic process PV^{γ} = constant. Here γ is the ratio of specific heats, P is the pressure and V is the volume of the ideal gas.

In the light of the above statements, choose the **correct** answer from the options given below:

(A) (A) is true but (R) is false

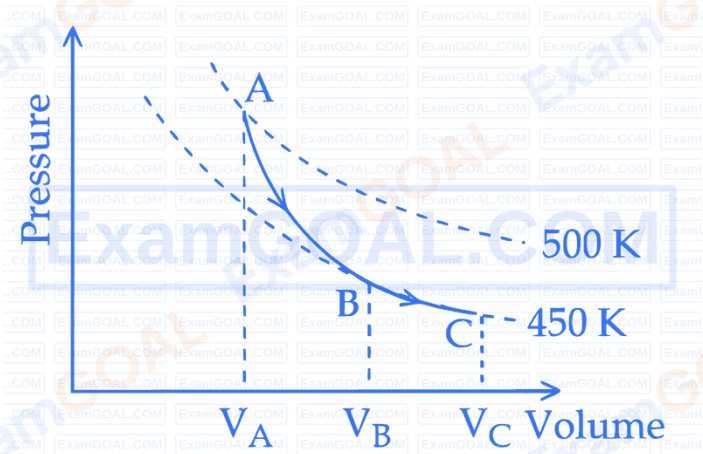
(B) Both (A) and (R) are true but (R) is NOT the correct explanation of (A)

(C) (A) is false but (R) is true

(D) Both (A) and (R) are true and (R) is the correct explanation of (A)

Q24. JEE Main 2025 (Online) 29th January Evening Shift

Open solution



A poly-atomic molecule ($C_V = 3R, C_P = 4R$, where R is gas constant) goes from phase space point A $\left(\mathrm{P_A} = 10^5\ \mathrm{Pa},\ \mathrm{V_A} = 4 \times 10^{-6}\ \mathrm{m^3}\right)$ to point B $\left(\mathrm{P_B} = 5 \times 10^4\ \mathrm{Pa},\ \mathrm{V_B} = 6 \times 10^{-6}\ \mathrm{m^3}\right)$ to point C $\left(\mathrm{P_C} = 10^4\ \mathrm{Pa}, \mathrm{V}_C = 8 \times 10^{-6}\ \mathrm{m^3}\right)$. A to B is an adiabatic path and B to C is an isothermal path.

The net heat absorbed per unit mole by the system is :

(A) $500\mathrm{R}(\ln 3 + \ln 4)$ (B) $450\mathrm{R}(\ln 4 - \ln 3)$ (C) $500\mathrm{R}\ln 2$ (D) $400\mathrm{R}\ln 4$

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Q25. JEE Main 2025 (Online) 29th January Morning Shift

The work done in an adiabatic change in an ideal gas depends upon only :

(A) change in its pressure

(B) change in its temperature

(C) change in its specific heat

(D) change in its volume

Open solution

Q26. JEE Main 2025 (Online) 28th January Morning Shift

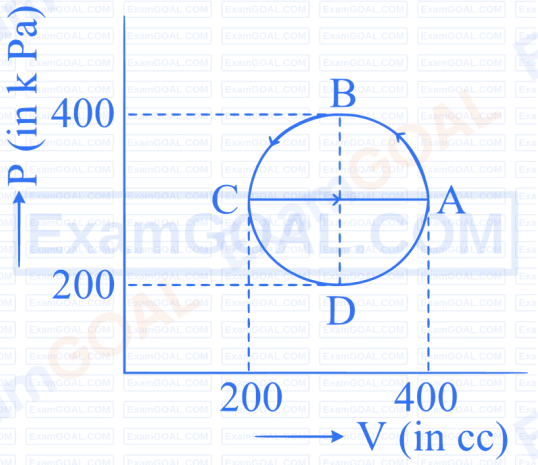
A Carnot engine (E) is working between two temperatures 473 K and 273 K . In a new system two engines - engine E_1 works between 473 K to 373 K and engine E_2 works between 373 K to 273 K . If η_{12}, η_1 and η_2 are the efficiencies of the engines E, E_1 and E_2, respectively, then

(A) $\eta_{12} = \eta_1\eta_2$ (B) $\eta_{12} = \eta_1 + \eta_2$ (C) $\eta_{12} \geq \eta_1 + \eta_2$ (D) $\eta_{12} < \eta_1 + \eta_2$

Open solution

Q27. JEE Main 2025 (Online) 24th January Evening Shift

The magnitude of heat exchanged by a system for the given cyclic process ABCA (as shown in figure) is (in SI unit) :



(A) zero

(B) 40π

(C) 5π

(D) 10π

Open solution

Q28. JEE Main 2025 (Online) 24th January Evening Shift

Given below are two statements. One is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A) : In an insulated container, a gas is adiabatically shrunk to half of its initial volume. The temperature of the gas decreases.

Reason (R) : Free expansion of an ideal gas is an irreversible and an adiabatic process.

In the light of the above statements, choose the correct answer from the options given below :

(A) Both $(\mathbf{A})$ and $(\mathbf{R})$ are true and $(\mathbf{R})$ is the correct explanation of $(\mathbf{A})$

(B) Both (A) and (R) are true but (R) is NOT the correct explanation of (A)

(C) (A) is false but (R) is true

(D) (A) is true but $(\mathbf{R})$ is false

Open solution

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Q29. JEE Main 2025 (Online) 24th January Morning Shift

Open solution

An ideal gas goes from an initial state to final state. During the process, the pressure of gas increases linearly with temperature.

A. The work done by gas during the process is zero.

B. The heat added to gas is different from change in its internal energy.

C. The volume of the gas is increased.

D. The internal energy of the gas is increased.

E. The process is isochoric (constant volume process)

Choose the correct answer from the options given below:

(A) A, C Only (B) A, D, E Only (C) E Only (D) A, B, C, D Only

Q30. JEE Main 2025 (Online) 23rd January Evening Shift

Open solution

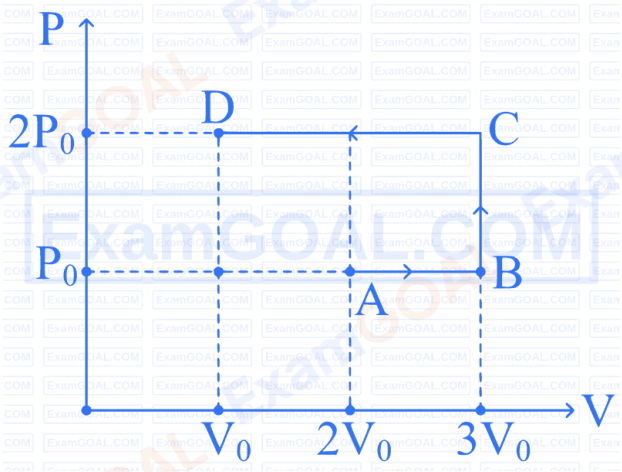
Water of mass m gram is slowly heated to increase the temperature from T_1 to T_2. The change in entropy of the water, given specific heat of water is $1 \mathrm{Jkg}^{-1} \mathrm{K}^{-1}$, is :

(A) $\mathrm{m}\left(\mathrm{T}_2-\mathrm{T}_1\right)$ (B) zero

(C) $\mathrm{m} \ln \left(\frac{\mathrm{T}_1}{\mathrm{T}_2}\right)$ (D) $\mathrm{m} \ln \left(\frac{\mathrm{T}_2}{\mathrm{T}_1}\right)$

Q31. JEE Main 2025 (Online) 23rd January Evening Shift

Open solution



Using the given $P-V$ diagram, the work done by an ideal gas along the path $ABCD$ is :

(A) $3 \mathrm{P}_0 \mathrm{V}_0$ (B) $-3 \mathrm{P}_0 \mathrm{V}_0$ (C) $-4 \mathrm{P}_0 \mathrm{V}_0$ (D) $4 \mathrm{P}_0 \mathrm{V}_0$

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Q32. JEE Main 2025 (Online) 23rd January Morning Shift

Open solution

Match the List - I with List - II

	List - I		List - II
(A)	Pressure varies inversely with volume of an ideal gas.	(I)	Adiabatic process
(B)	Heat absorbed goes partly to increase internal energy and partly to do work.	(II)	Isochoric process
(C)	Heat is neither absorbed nor released by a system.	(III)	Isothermal process
(D)	No work is done on or by a gas.	(IV)	Isobaric process

Choose the correct answer from the options given below:

(A) A-I, B-III, C-II, D-IV

(B) A-I, B-IV, C-II, D-III

(C) A-III, B-I, C-IV, D-II

(D) A-III, B-IV, C-I, D-II

Q33. JEE Main 2025 (Online) 22nd January Evening Shift

Open solution

Given are statements for certain thermodynamic variables,

(A) Internal energy, volume (V) and mass (M) are extensive variables.

(B) Pressure (P), temperature (T) and density (ρ) are intensive variables.

(C) Volume (V), temperature (T) and density (ρ) are intensive variables.

(D) Mass (M), temperature (T) and internal energy are extensive variables.

Choose the correct answer from the options given below :

(A) (C) and (D) Only

(B) (A) and (B) Only

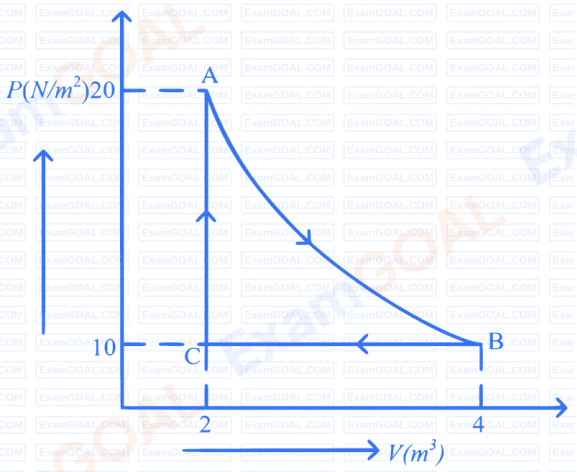
(C) (D) and (A) Only

(D) (B) and (C) Only

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Q34. JEE Main 2024 (Online) 9th April Evening Shift

A real gas within a closed chamber at $27^{\circ}\mathrm{C}$ undergoes the cyclic process as shown in figure. The gas obeys $PV^3 = RT$ equation for the path A to B. The net work done in the complete cycle is (assuming $R = 8\ \mathrm{J/molK}$):



(A) $-20\mathrm{J}$ (B) 205J (C) 225J (D) 20J

Open solution

Q35. JEE Main 2024 (Online) 9th April Morning Shift

The volume of an ideal gas $(\gamma = 1.5)$ is changed adiabatically from 5 litres to 4 litres. The ratio of initial pressure to final pressure is :

(A) $\frac{4}{5}$ (B) $\frac{8}{5\sqrt{5}}$ (C) $\frac{2}{\sqrt{5}}$ (D) $\frac{16}{25}$

Open solution

Q36. JEE Main 2024 (Online) 9th April Morning Shift

A sample of 1 mole gas at temperature T is adiabatically expanded to double its volume. If adiabatic constant for the gas is $\gamma = \frac{3}{2}$, then the work done by the gas in the process is :

(A) $\mathrm{RT}[2+\sqrt{2}]$ (B) $\mathrm{RT}[2-\sqrt{2}]$
(C) $\frac{\mathrm{R}}{\mathrm{T}}[2-\sqrt{2}]$ (D) $\frac{T}{R}[2+\sqrt{2}]$

Open solution

Q37. JEE Main 2024 (Online) 8th April Evening Shift

A diatomic gas $(\gamma = 1.4)$ does $100\ \mathrm{J}$ of work in an isobaric expansion. The heat given to the gas is :

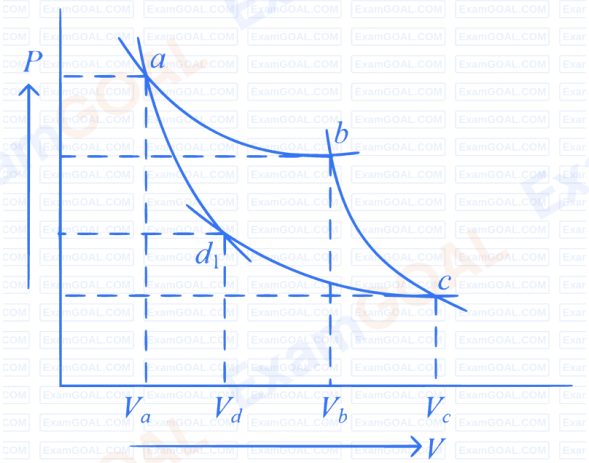
(A) 150 J (B) 490 J (C) 350 J (D) 250 J

Open solution

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Q38. JEE Main 2024 (Online) 8th April Morning Shift

Two different adiabatic paths for the same gas intersect two isothermal curves as shown in P-V diagram. The relation between the ratio $\frac{V_a}{V_d}$ and the ratio $\frac{V_b}{V_c}$ is:



Open solution

(A) $\frac{V_a}{V_d} \neq \frac{V_b}{V_c}$

(B) $\frac{V_a}{V_d} = \left(\frac{V_b}{V_c}\right)^{-1}$

(C) $\frac{V_a}{V_d} = \frac{V_b}{V_c}$

(D) $\frac{V_a}{V_d} = \left(\frac{V_b}{V_c}\right)^{2}$

Q39. JEE Main 2024 (Online) 6th April Evening Shift

A total of $48\ \mathrm{J}$ heat is given to one mole of helium kept in a cylinder. The temperature of helium increases by $2^{\circ}\mathrm{C}$. The work done by the gas is: Given, $\mathrm{R} = 8.3\ \mathrm{J\ K^{-1}\ mol^{-1}}$.

(A) 23.1 J (B) 48 J (C) 24.9 J (D) 72.9 J

Open solution

Q40. JEE Main 2024 (Online) 6th April Morning Shift

The specific heat at constant pressure of a real gas obeying $PV^2 = RT$ equation is:

(A) R (B) $C_V + R$ (C) $C_V + \frac{R}{2V}$ (D) $\frac{R}{3} + C_V$

Open solution

Q41. JEE Main 2024 (Online) 5th April Evening Shift

During an adiabatic process, if the pressure of a gas is found to be proportional to the cube of its absolute temperature, then the ratio of $\frac{\mathrm{C_P}}{\mathrm{C_V}}$ for the gas is :

(A) $\frac{5}{3}$ (B) $\frac{3}{2}$ (C) $\frac{7}{5}$ (D) $\frac{9}{7}$

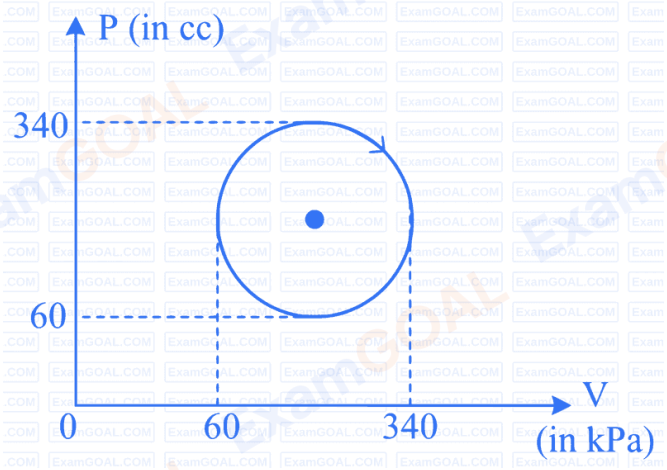
Open solution

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Q42. JEE Main 2024 (Online) 5th April Morning Shift

Open solution

The heat absorbed by a system in going through the given cyclic process is :



(A) 61.6 J (B) 431.2 J (C) 19.6 J (D) 616 J

Q43. JEE Main 2024 (Online) 4th April Evening Shift

Open solution

A sample of gas at temperature T is adiabatically expanded to double its volume. Adiabatic constant for the gas is $\gamma = 3/2$. The work done by the gas in the process is:

$(\mu = 1$ mole $)$

(A) $RT[2\sqrt{2} - 1]$ (B) $RT[2 - \sqrt{2}]$ (C) $RT[1 - 2\sqrt{2}]$ (D) $RT[\sqrt{2} - 2]$

Q44. JEE Main 2024 (Online) 1st February Evening Shift

Open solution

A diatomic gas $(\gamma = 1.4)$ does 200 J of work when it is expanded isobarically. The heat given to the gas in the process is :

(A) 800 J (B) 600 J (C) 700 J (D) 850 J

Q45. JEE Main 2024 (Online) 1st February Morning Shift

Open solution

The pressure and volume of an ideal gas are related as $\mathrm{PV}^{\frac{3}{2}} = \mathrm{K}$ (Constant). The work done when the gas is taken from state $A\,(P_1, V_1, T_1)$ to state $B\,(P_2, V_2, T_2)$ is :

(A) $2\left(\mathrm{P}_2\sqrt{\mathrm{V}_2} - \mathrm{P}_1\sqrt{\mathrm{V}_1}\right)$

(B) $2\left(\sqrt{\mathrm{P}_1}\,\mathrm{V}_1 - \sqrt{\mathrm{P}_2}\,\mathrm{V}_2\right)$

(C) $2\left(\mathrm{P}_2\,\mathrm{V}_2 - \mathrm{P}_1\,\mathrm{V}_1\right)$

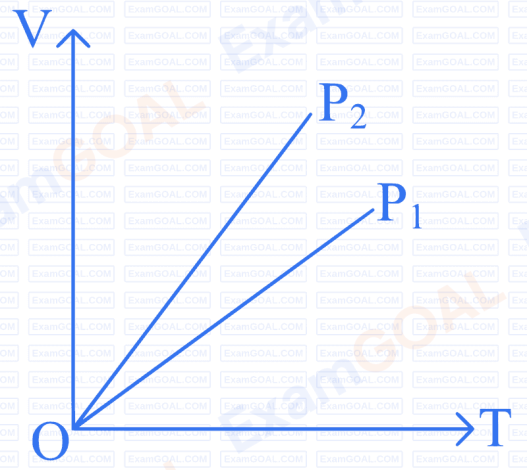
(D) $2\left(\mathrm{P}_1\,\mathrm{V}_1 - \mathrm{P}_2\,\mathrm{V}_2\right)$

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Q46. JEE Main 2024 (Online) 31st January Morning Shift

Open solution

The given figure represents two isobaric processes for the same mass of an ideal gas, then

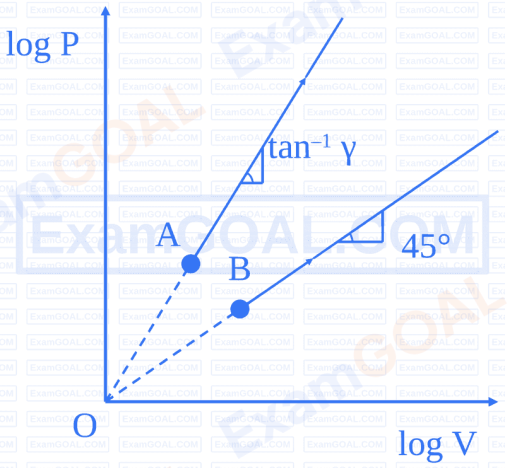


(A) $P_2 > P_1$ (B) $P_1 > P_2$ (C) $P_1 = P_2$ (D) $P_2 \geq P_1$

Q47. JEE Main 2024 (Online) 30th January Morning Shift

Open solution

Two thermodynamical processes are shown in the figure. The molar heat capacity for process A and B are C_A and C_B. The molar heat capacity at constant pressure and constant volume are represented by C_P and C_V, respectively. Choose the correct statement.



(A) $C_P > C_B > C_A > C_V$

(B) $C_P > C_V > C_A = C_B$

(C) $C_A = 0$ and $C_B = \infty$

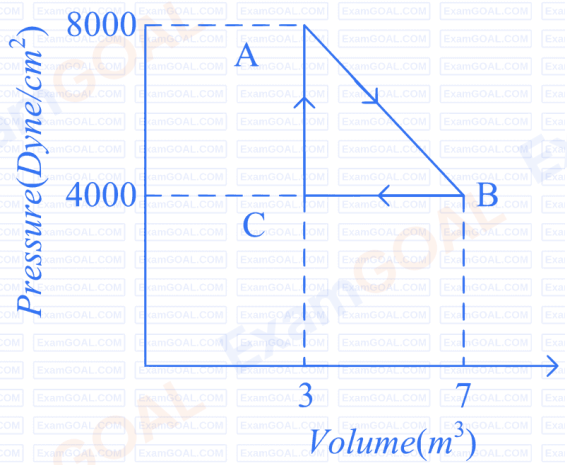
(D) $C_A = \infty, C_B = 0$

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Q48. JEE Main 2024 (Online) 29th January Morning Shift

Open solution

A thermodynamic system is taken from an original state A to an intermediate state B by a linear process as shown in the figure. Its volume is then reduced to the original value from B to C by an isobaric process. The total work done by the gas from A to B and B to C would be :



(A) 800 J (B) 2200 J (C) 33800 J (D) 1200 J

Q49. JEE Main 2024 (Online) 27th January Evening Shift

Open solution

During an adiabatic process, the pressure of a gas is found to be proportional to the cube of its absolute temperature. The ratio of $\frac{\mathrm{Cp}}{\mathrm{Cv}}$ for the gas is :

(A) $\frac{7}{5}$ (B) $\frac{3}{2}$ (C) $\frac{9}{7}$ (D) $\frac{5}{3}$

Q50. JEE Main 2024 (Online) 27th January Morning Shift

Open solution

$0.08\ \mathrm{kg}$ air is heated at constant volume through $5^{\circ}\mathrm{C}$. The specific heat of air at constant volume is $0.17\ \mathrm{kcal/kg^{\circ}C}$ and $\mathrm{J} = 4.18$ joule/ cal. The change in its internal energy is approximately.

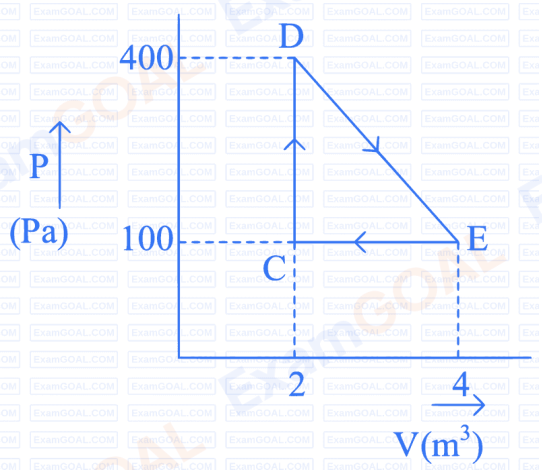
(A) 318 J (B) 298 J (C) 284 J (D) 142 J

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Q51. JEE Main 2023 (Online) 15th April Morning Shift

A thermodynamic system is taken through cyclic process. The total work done in the process is :

Open solution



(A) $100\ \text{J}$ (B) Zero (C) $300\ \text{J}$ (D) $200\ \text{J}$

Q52. JEE Main 2023 (Online) 13th April Evening Shift

The initial pressure and volume of an ideal gas are P_0 and V_0. The final pressure of the gas when the gas is suddenly compressed to volume $\frac{V_0}{4}$ will be :

(Given γ = ratio of specific heats at constant pressure and at constant volume)

Open solution

(A) $P_0(4)^{\frac{1}{\gamma}}$ (B) P_0 (C) $4P_0$ (D) $P_0(4)^{\gamma}$

Q53. JEE Main 2023 (Online) 11th April Evening Shift

The Thermodynamic process, in which internal energy of the system remains constant is

Open solution

(A) Isobaric (B) Isochoric (C) Adiabatic (D) Isothermal

Q54. JEE Main 2023 (Online) 11th April Morning Shift

$1\ \text{kg}$ of water at 100°C is converted into steam at 100°C by boiling at atmospheric pressure. The volume of water changes from $1.00 \times 10^{-3}\ \text{m}^3$ as a liquid to $1.671\ \text{m}^3$ as steam. The change in internal energy of the system during the process will be

Open solution

(Given latent heat of vaporisation $= 2257\ \text{kJ/kg}$, Atmospheric pressure = $1 \times 10^5\ \text{Pa}$)

(A) + 2090 kJ (B) $-$ 2426 kJ (C) + 2476 kJ (D) $-$ 2090 kJ

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Q55. JEE Main 2023 (Online) 10th April Evening Shift

A gas is compressed adiabatically, which one of the following statements is NOT true?

(A) There is no heat supplied to the system

(B) The temperature of the gas increases.

(C) There is no change in the internal energy

(D) The change in the internal energy is equal to the work done on the gas.

Open solution

Q56. JEE Main 2023 (Online) 10th April Morning Shift

Consider two containers A and B containing monoatomic gases at the same Pressure (P), Volume (V) and Temperature (T). The gas in A is compressed isothermally to $\frac{1}{8}$ of its original volume while the gas in B is compressed adiabatically to $\frac{1}{8}$ of its original volume. The ratio of final pressure of gas in B to that of gas in A is

(A) $\frac{1}{8}$ (B) $8^{\frac{3}{2}}$ (C) 4 (D) 8

Open solution

Q57. JEE Main 2023 (Online) 8th April Morning Shift

Given below are two statements:

Statement I: If heat is added to a system, its temperature must increase.

Statement II: If positive work is done by a system in a thermodynamic process, its volume must increase.

In the light of the above statements, choose the correct answer from the options given below

(A) Both Statement I and Statement II are true

(B) Statement I is false but Statement II is true

(C) Statement I is true but Statement II is false

(D) Both Statement I and Statement II are false

Open solution

Q58. JEE Main 2023 (Online) 6th April Morning Shift

A source supplies heat to a system at the rate of $1000\ \mathrm{W}$. If the system performs work at a rate of $200\ \mathrm{W}$. The rate at which internal energy of the system increases is

(A) 600 W (B) 1200 W (C) 500 W (D) 800 W

Open solution

Q59. JEE Main 2023 (Online) 1st February Morning Shift

A sample of gas at temperature T is adiabatically expanded to double its volume. The work done by the gas in the process is $\left(\text{given}, \gamma = \frac{3}{2}\right)$:

(A) $W = TR[\sqrt{2} - 2]$ (B) $W = \frac{T}{R}[\sqrt{2} - 2]$ (C) $W = \frac{R}{T}[2 - \sqrt{2}]$ (D) $W = RT[2 - \sqrt{2}]$

Open solution

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Q60. JEE Main 2023 (Online) 31st January Evening Shift

Heat energy of $735\ \text{J}$ is given to a diatomic gas allowing the gas to expand at constant pressure. Each gas molecule rotates around an internal axis but do not oscillate. The increase in the internal energy of the gas will be :

Open solution

(A) $572\ \text{J}$ (B) $441\ \text{J}$ (C) $525\ \text{J}$ (D) $735\ \text{J}$

Q61. JEE Main 2023 (Online) 31st January Evening Shift

A hypothetical gas expands adiabatically such that its volume changes from 8 litres to 27 litres. If the ratio of final pressure of the gas to initial pressure of the gas is $\frac{16}{81}$. Then the ratio of $\frac{C_p}{C_v}$ will be.

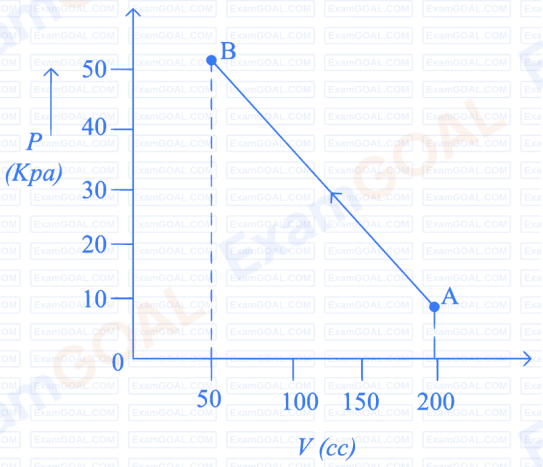
Open solution

(A) $\frac{3}{1}$ (B) $\frac{4}{3}$ (C) $\frac{1}{2}$ (D) $\frac{3}{2}$

Q62. JEE Main 2023 (Online) 31st January Morning Shift

The pressure of a gas changes linearly with volume from A to B as shown in figure. If no heat is supplied to or extracted from the gas then change in the internal energy of the gas will be

Open solution



(A) 6 J (B) 4.5 J (C) zero (D) -4.5 J

Q63. JEE Main 2023 (Online) 30th January Morning Shift

Heat is given to an ideal gas in an isothermal process.

A. Internal energy of the gas will decrease.

B. Internal energy of the gas will increase.

C. Internal energy of the gas will not change.

D. The gas will do positive work.

E. The gas will do negative work.

Choose the correct answer from the options given below :

Open solution

(A) B and D only (B) C and E only (C) A and E only (D) C and D only

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Q64. JEE Main 2023 (Online) 29th January Morning Shift

Open solution

Given below are two statements: One is labelled as Assertion A and the other is labelled as Reason R.

Assertion A: If dQ and dW represent the heat supplied to the system and the work done on the system respectively. Then according to the first law of thermodynamics $dQ = dU - dW$.

Reason R: First law of thermodynamics is based on law of conservation of energy.

In the light of the above statements, choose the correct answer from the options given below:

(A) Both A and R are correct but R is not the correct explanation of A

(B) Both A and R are correct and R is the correct explanation of A

(C) A is correct but R is not correct

(D) A is not correct but R is correct

Q65. JEE Main 2023 (Online) 25th January Evening Shift

Open solution

Match List I with List II

	List I		List II
A.	Isothermal Process	I.	Work done by the gas decreases internal energy
B.	Adiabatic Process	II.	No change in internal energy
C.	Isochoric Process	III.	The heat absorbed goes partly to increase internal energy and partly to do work
D.	Isobaric Process	IV.	No work is done on or by the gas

Choose the correct answer from the options given below :

(A) A-I, B-II, C-IV, D-III

(B) A-II, B-I, C-III, D-IV

(C) A-II, B-I, C-IV, D-III

(D) A-I, B-II, C-III, D-IV

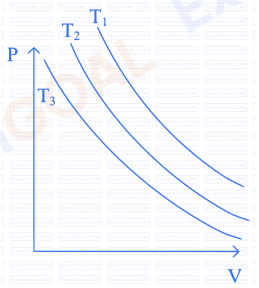
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Q66. JEE Main 2023 (Online) 24th January Evening Shift

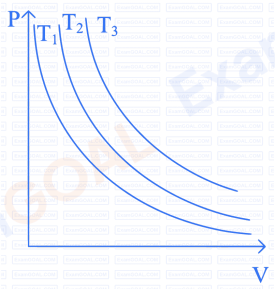
Open solution

In an Isothermal change, the change in pressure and volume of a gas can be represented for three different temperatures; $T_3 > T_2 > T_1$ as :

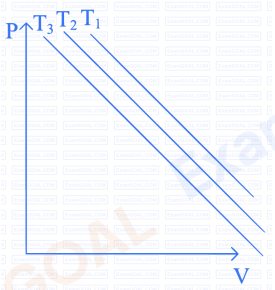
(A)



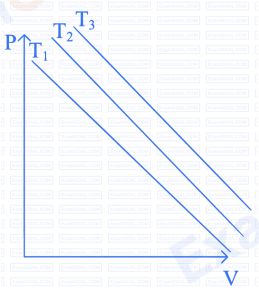
(B)



(C)



(D)



Q67. JEE Main 2023 (Online) 24th January Morning Shift

Open solution

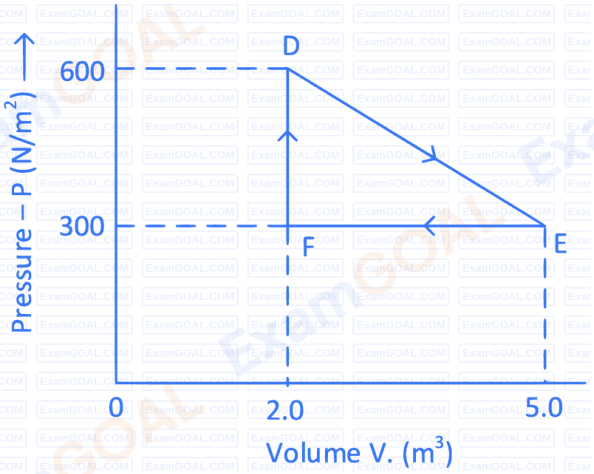
1 g of a liquid is converted to vapour at 3×10^5 Pa pressure. If 10% of the heat supplied is used for increasing the volume by 1600 cm^3 during this phase change, then the increase in internal energy in the process will be :

(A) 4800 J (B) 4320 J (C) 432000 J (D) 4.32×10^8 J

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Q68. JEE Main 2022 (Online) 29th July Evening Shift

A thermodynamic system is taken from an original state D to an intermediate state E by the linear process shown in the figure. Its volume is then reduced to the original volume from E to F by an isobaric process. The total work done by the gas from D to E to F will be



Open solution

(A) -450 J (B) 450 J (C) 900 J (D) 1350 J

Q69. JEE Main 2022 (Online) 28th July Evening Shift

A vessel contains $14\ \mathrm{g}$ of nitrogen gas at a temperature of $27^{\circ}\mathrm{C}$. The amount of heat to be transferred to the gas to double the r.m.s speed of its molecules will be :

Take $\mathrm{R} = 8.32\ \mathrm{J\ mol^{-1}\ k^{-1}}$.

Open solution

(A) 2229 J (B) 5616 J (C) 9360 J (D) 13,104 J

Q70. JEE Main 2022 (Online) 26th July Morning Shift

7 mol of a certain monoatomic ideal gas undergoes a temperature increase of $40\ \mathrm{K}$ at constant pressure. The increase in the internal energy of the gas in this process is :

(Given $\mathrm{R} = 8.3\ \mathrm{JK^{-1}\ mol^{-1}}$)

Open solution

(A) 5810 J (B) 3486 J (C) 11620 J (D) 6972 J

Q71. JEE Main 2022 (Online) 26th July Morning Shift

A monoatomic gas at pressure P and volume V is suddenly compressed to one eighth of its original volume. The final pressure at constant entropy will be :

Open solution

(A) P (B) 8P (C) 32P (D) 64P

Q72. JEE Main 2022 (Online) 25th July Morning Shift

A certain amount of gas of volume V at $27^{\circ}\mathrm{C}$ temperature and pressure $2 \times 10^{7}\mathrm{Nm^{-2}}$ expands isothermally until its volume gets doubled. Later it expands adiabatically until its volume gets redoubled. The final pressure of the gas will be (Use $\gamma = 1.5$) :

Open solution

(A) $3.536 \times 10^{5}\ \mathrm{Pa}$ (B) $3.536 \times 10^{6}\ \mathrm{Pa}$ (C) $1.25 \times 10^{6}\ \mathrm{Pa}$ (D) $1.25 \times 10^{5}\ \mathrm{Pa}$

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Q73. JEE Main 2022 (Online) 30th June Morning Shift

A sample of monoatomic gas is taken at initial pressure of 75 kPa. The volume of the gas is then compressed from 1200 cm^3 to 150 cm^3 adiabatically. In this process, the value of workdone on the gas will be :

Open solution

(A) 79 J (B) 405 J (C) 4050 J (D) 9590 J

Q74. JEE Main 2022 (Online) 29th June Evening Shift

Starting with the same initial conditions, an ideal gas expands from volume V_1 to V_2 in three different ways. The work done by the gas is W_1 if the process is purely isothermal, W_2, if the process is purely adiabatic and W_3 if the process is purely isobaric. Then, choose the correct option

Open solution

(A) $W_1 < W_2 < W_3$ (B) $W_2 < W_3 < W_1$ (C) $W_3 < W_1 < W_2$ (D) $W_2 < W_1 < W_3$

Q75. JEE Main 2022 (Online) 29th June Morning Shift

A cylinder of fixed capacity of 44.8 litres contains helium gas at standard temperature and pressure. The amount of heat needed to raise the temperature of gas in the cylinder by 20.0°C will be :

(Given gas constant R = 8.3 JK^{-1}-mol^{-1})

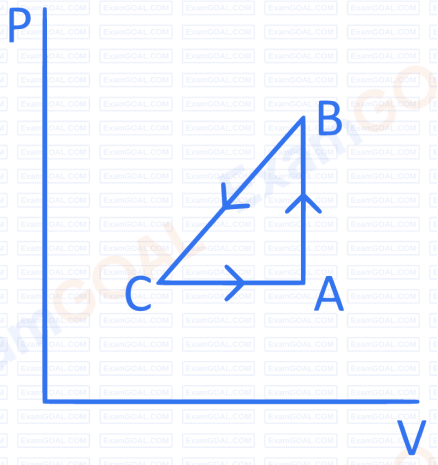
Open solution

(A) 249 J (B) 415 J (C) 498 J (D) 830 J

Q76. JEE Main 2022 (Online) 28th June Evening Shift

A sample of an ideal gas is taken through the cyclic process ABCA as shown in figure. It absorbs, 40 J of heat during the part AB, no heat during BC and rejects 60 J of heat during CA. A work of 50 J is done on the gas during the part BC. The internal energy of the gas at A is 1560 J. The workdone by the gas during the part CA is :

Open solution



(A) 20 J (B) 30 J (C) −30 J (D) −60 J

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Q77. JEE Main 2022 (Online) 28th June Morning Shift

Open solution

Given below are two statements :

Statement I : When μ amount of an ideal gas undergoes adiabatic change from state (P_1, V_1, T_1) to state (P_2, V_2, T_2), then work done is $W = \frac{\mu R(T_2 - T_1)}{1-\gamma}$, where $\gamma = \frac{C_p}{C_v}$ and R = universal gas constant.

Statement II : In the above case, when work is done on the gas, the temperature of the gas would rise.

Choose the correct answer from the options given below :

(A) Both Statement I and Statement II are true.

(B) Both Statement I and Statement II are false.

(C) Statement I is true but Statement II is false.

(D) Statement I is false but Statement II is true.

Q78. JEE Main 2022 (Online) 26th June Morning Shift

Open solution

A thermally insulated vessel contains an ideal gas of molecular mass M and ratio of specific heats 1.4. Vessel is moving with speed v and is suddenly brought to rest. Assuming no heat is lost to the surrounding and vessel, temperature of the gas increases by :

(R = universal gas constant)

(A) $\frac{Mv^2}{7R}$ (B) $\frac{Mv^2}{5R}$ (C) $2\frac{Mv^2}{7R}$ (D) $7\frac{Mv^2}{5R}$

Q79. JEE Main 2021 (Online) 31st August Morning Shift

Open solution

For an ideal gas the instantaneous change in pressure 'p' with volume 'v' is given by the equation $\frac{dp}{dv} = -ap$. If p = p_0 at v =0 is the given boundary condition, then the maximum temperature one mole of gas can attain is : (Here R is the gas constant)

(A) $\frac{p_0}{aeR}$ (B) $\frac{ap_0}{eR}$ (C) infinity (D) 0°C

Q80. JEE Main 2021 (Online) 26th August Morning Shift

Open solution

An electric appliance supplies 6000 J/min heat to the system. If the system delivers a power of 90W. How long it would take to increase the internal energy by 2.5×10^3 J ?

(A) 2.5×10^2 s (B) 4.1×10^1 s (C) 2.4×10^3 s (D) 2.5×10^1 s

Q81. JEE Main 2021 (Online) 27th July Evening Shift

Open solution

One mole of an ideal gas is taken through an adiabatic process where the temperature rises from 27° C to 37° C. If the ideal gas is composed of polyatomic molecule that has 4 vibrational modes, which of the following is true? [R = 8.314 J mol^{-1} K^{-1}]

(A) work done by the gas is close to 332 J (B) work done on the gas is close to 582 J

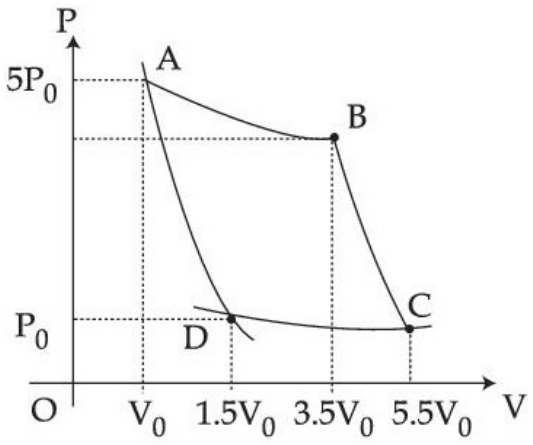
(C) work done by the gas is close to 582 J (D) work done on the gas is close to 332 J

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Q82. JEE Main 2021 (Online) 27th July Morning Shift

Open solution

In the reported figure, there is a cyclic process ABCDA on a sample of 1 mol of a diatomic gas. The temperature of the gas during the process A $\rightarrow$ B and C $\rightarrow$ D are T_1 and T_2 ($T_1 > T_2$) respectively.



Choose the correct option out of the following for work done if processes BC and DA are adiabatic.

(A) $W_{AB} = W_{DC}$ (B) $W_{AD} = W_{BC}$ (C) $W_{BC} + W_{DA} > 0$ (D) $W_{AB} < W_{CD}$

Q83. JEE Main 2021 (Online) 25th July Morning Shift

Open solution

For a gas $C_P - C_V = R$ in a state P and $C_P - C_V = 1.10\ R$ in a state Q, T_P and T_Q are the temperatures in two different states P and Q respectively. Then

(A) $T_P = T_Q$ (B) $T_P < T_Q$ (C) $T_P = 0.9\ T_Q$ (D) $T_P > T_Q$

Q84. JEE Main 2021 (Online) 25th July Morning Shift

Open solution

A monoatomic ideal gas, initially at temperature T_1 is enclosed in a cylinder fitted with a frictionless piston. The gas is allowed to expand adiabatically to a temperature T_2 by releasing the piston suddenly. If l_1 and l_2 are the lengths of the gas column, before and after the expansion respectively, then the value of $\frac{T_1}{T_2}$ will be :

(A) $\left(\frac{l_1}{l_2}\right)^{\frac{2}{3}}$ (B) $\left(\frac{l_2}{l_1}\right)^{\frac{2}{3}}$

(C) $\frac{l_2}{l_1}$ (D) $\frac{l_1}{l_2}$

Q85. JEE Main 2021 (Online) 20th July Morning Shift

Open solution

The amount of heat needed to raise the temperature of 4 moles of rigid diatomic gas from 0° C to 50° C when no work is done is _____. (R is the universal gas constant).

(A) 500 R (B) 250 R (C) 750 R (D) 175 R

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Q86. JEE Main 2021 (Online) 18th March Evening Shift

Open solution

For an adiabatic expansion of an ideal gas, the fractional change in its pressure is equal to (where γ is the ratio of specific heats) :

(A) $-\frac{1}{\gamma}\frac{dV}{V}$

(B) $-\gamma\frac{V}{dV}$

(C) $-\gamma\frac{dV}{V}$

(D) $\frac{dV}{V}$

Q87. JEE Main 2021 (Online) 18th March Evening Shift

Open solution

An ideal gas in a cylinder is separated by a piston in such a way that the entropy of one part is S_1 and that of the other part is S_2. Given that $S_1 > S_2$. If the piston is removed then the total entropy of the system will be :

(A) $S_1 - S_2$

(B) $\frac{S_1}{S_2}$

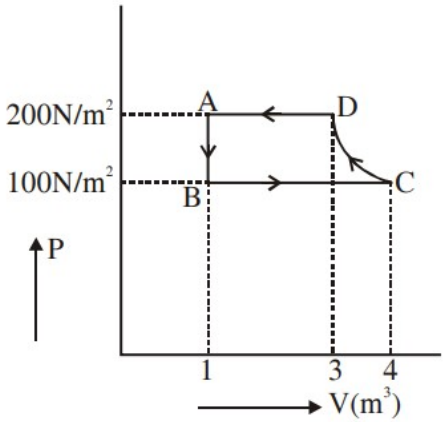
(C) $S_1 \times S_2$

(D) $S_1 + S_2$

Q88. JEE Main 2021 (Online) 18th March Morning Shift

Open solution

The P-V diagram of a diatomic ideal gas system going under cyclic process as shown in figure. The work done during an adiabatic process CD is (use γ = 1.4) :



(A) −500 J

(B) −400 J

(C) 400 J

(D) 200 J

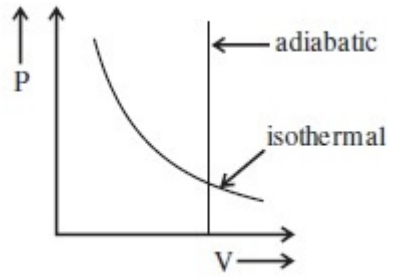
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Q89. JEE Main 2021 (Online) 17th March Evening Shift

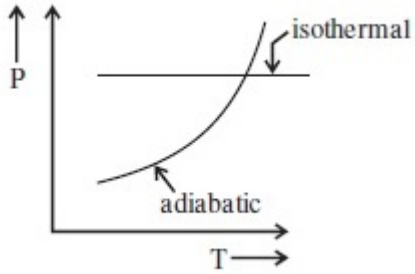
Open solution

Which one is the correct option for the two different thermodynamic processes?

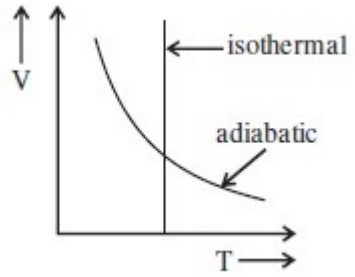
(a)



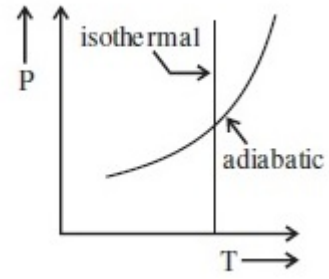
(b)



(c)



(d)



(A) (a) only **(B)** (c) and (d) **(C)** (b) and (c) **(D)** (c) and (a)

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Q90. JEE Main 2021 (Online) 16th March Morning Shift

In thermodynamics, heat and work are :

(A) Path functions

(B) Point functions

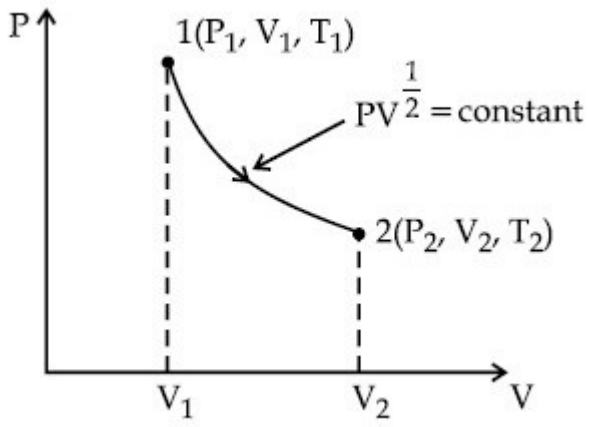
(C) Extensive thermodynamics state variables

(D) Intensive thermodynamic state variables

Open solution

Q91. JEE Main 2021 (Online) 25th February Evening Shift

Thermodynamic process is shown below on a P-V diagram for one mole of an ideal gas. If $V_2 = 2V_1$ then the ratio of temperature T_2/T_1 is :



(A) $\sqrt{2}$

(B) $\frac{1}{\sqrt{2}}$

(C) $\frac{1}{2}$

(D) 2

Open solution

Q92. JEE Main 2021 (Online) 25th February Morning Shift

A diatomic gas, having $C_p = \frac{7}{2}R$ and $C_v = \frac{5}{2}R$, is heated at constant pressure. The ratio dU : dQ : dW :

(A) 5 : 7 : 3

(B) 3 : 7 : 2

(C) 5 : 7 : 2

(D) 3 : 5 : 2

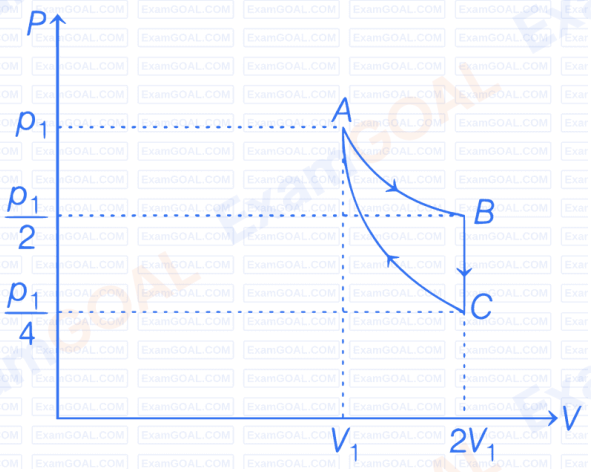
Open solution

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Q93. JEE Main 2021 (Online) 24th February Evening Shift

Open solution

If one mole of an ideal gas at (P_1, V_1) is allowed to expand reversibly and isothermally (A to B) its pressure is reduced to one-half of the original pressure (see figure). This is followed by a constant volume cooling till its pressure is reduced to one-fourth of the initial value (B $\rightarrow$ C). Then it is restored to its initial state by a reversible adiabatic compression (C to A). The net work done by the gas is equal to :



(A) $-\frac{RT}{2(\gamma-1)}$

(B) $RT\left(\ln 2-\frac{1}{2(\gamma-1)}\right)$

(C) $RT\ln 2$

(D) 0

Q94. JEE Main 2021 (Online) 24th February Morning Shift

Open solution

Match List I with List II.

	List I		List II
(a)	Isothermal	(i)	Pressure constant
(b)	Isochoric	(ii)	Temperature constant
(c)	Adiabatic	(iii)	Volume constant
(d)	Isobaric	(iv)	Heat content is constant

Choose the correct answer from the options given below :

(1) (a) - (ii), (b) - (iii), (c) - (iv), (d) - (i)

(2) (a) - (ii), (b) - (iv), (c) - (iii), (d) - (i)

(3) (a) - (iii), (b) - (ii), (c) - (i), (d) - (iv)

(4) (a) - (i), (b) - (iii), (c) - (ii), (d) - (iv)

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Q95. JEE Main 2021 (Online) 24th February Morning Shift

Open solution

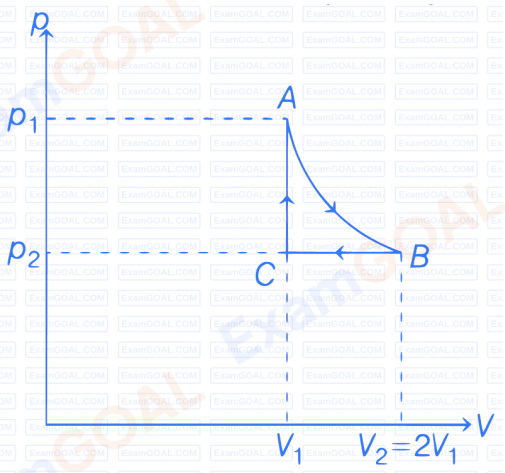
n mole of a perfect gas undergoes a cyclic process ABCA (see figure) consisting of the following processes.

A $\rightarrow$ B : Isothermal expansion at temperature T so that the volume is doubled from V_1 to $V_2 = 2V_1$ and pressure changes from P_1 to P_2

B $\rightarrow$ C : Isobaric compression at pressure P_2 to initial volume V_1.

C $\rightarrow$ A : Isochoric change leading to change of pressure from P_2 to P_1.

Total work done in the complete cycle ABCA is :



(A) nRTln 2

(B) 0

(C) $nRT\left(\ln 2 - \frac{1}{2}\right)$

(D) $nRT\left(\ln 2 + \frac{1}{2}\right)$

Q96. JEE Main 2020 (Online) 5th September Evening Slot

Open solution

In an adiabatic process, the density of a diatomic gas becomes 32 times its initial value. The final pressure of the gas is found to be n times the initial pressure. The value of n is :

(A) 128

(B) 32

(C) 326

(D) $\frac{1}{32}$

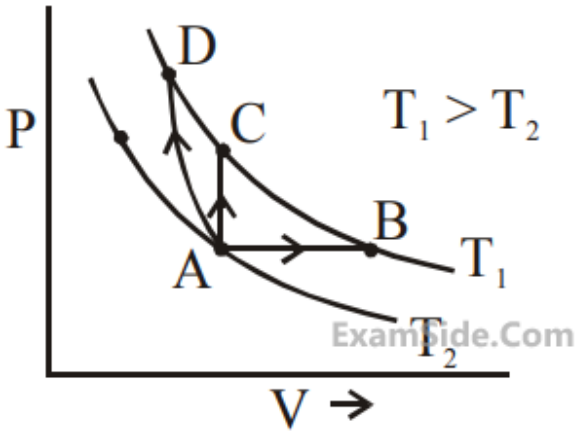
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Q97. JEE Main 2020 (Online) 5th September Morning Slot

Open solution

Three different processes that can occur in an ideal monoatomic gas are shown in the P vs V diagram. The paths are labelled as A $\rightarrow$ B, A $\rightarrow$ C and A $\rightarrow$ D. The change in internal energies during these process are taken as E_{AB}, E_{AC} and E_{AD} and the work done as W_{AB}, W_{AC} and W_{AD}.

The correct relation between these parameters are :



(A) $E_{AB} < E_{AC} < E_{AD}$, $W_{AB} > 0$, $W_{AC} > W_{AD}$

(B) $E_{AB} = E_{AC} = E_{AD}$, $W_{AB} > 0$, $W_{AC} = 0$, $W_{AD} < 0$

(C) $E_{AB} > E_{AC} > E_{AD}$, $W_{AB} < W_{AC} < W_{AD}$

(D) $E_{AB} = E_{AC} < E_{AD}$, $W_{AB} > 0$, $W_{AC} = 0$, $W_{AD} < 0$

Q98. JEE Main 2020 (Online) 4th September Evening Slot

Open solution

Match the thermodynamic processes taking place in a system with the correct conditions. In the table : ΔQ is the heat supplied, ΔW is the work done and ΔU is change in internal energy of the system.

Process	Condition
(I) Adiabatic	(1) $\Delta W = 0$
(II) Isothermal	(2) $\Delta Q = 0$
(III) Isochoric	(3) $\Delta U \neq 0$, $\Delta W \neq 0$, $\Delta Q \neq 0$
(IV) Isobaric	(4) $\Delta U = 0$

(A) (I) - (1), (II) - (1), (III) - (2), (IV) - (3)

(B) (I) - (2), (II) - (4), (III) - (1), (IV) - (3)

(C) (I) - (1), (II) - (2), (III) - (4), (IV) - (4)

(D) (I) - (2), (II) - (1), (III) - (4), (IV) - (3)

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Q99. JEE Main 2020 (Online) 3rd September Morning Slot

A balloon filled with helium (32°C and 1.7 atm.) bursts. Immediately afterwards the expansion of helium can be considered as

Open solution

(A) Irreversible adiabatic

(B) Reversible adiabatic

(C) Irreversible isothermal

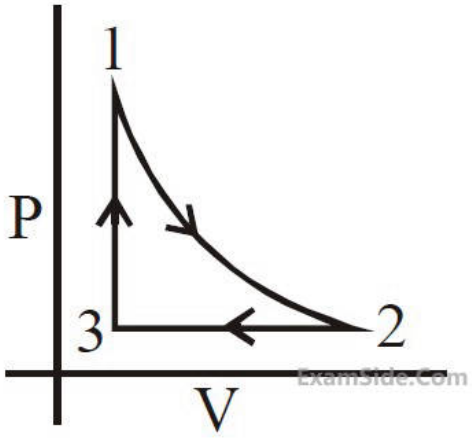
(D) Reversible isothermal

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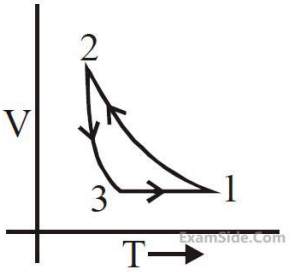
Q100. JEE Main 2020 (Online) 9th January Morning Slot

Open solution

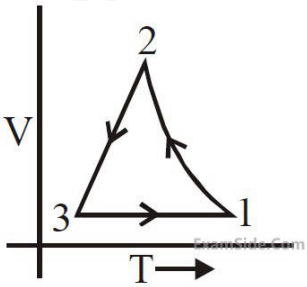
Which of the following is an equivalent cyclic process corresponding to the thermodynamic cycle given in the figure ? where, $1 \rightarrow 2$ is adiabatic.
(Graphs are schematic and are not to scale)



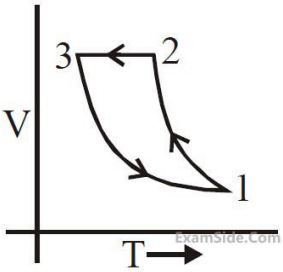
(A)



(B)

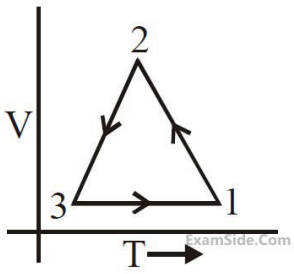


(C)



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(D)

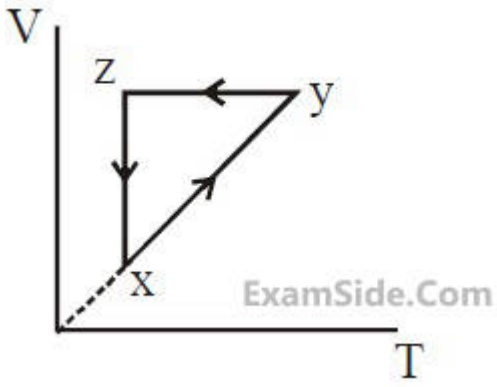


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Q101. JEE Main 2020 (Online) 8th January Morning Slot

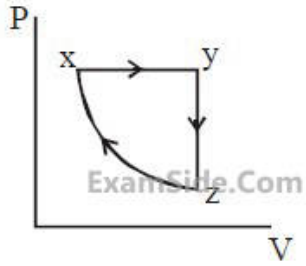
Open solution

A thermodynamic cycle xyzx is shown on a V-T diagram.

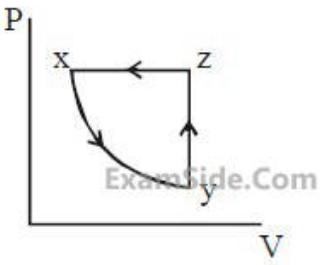


The P-V diagram that best describes this cycle is :
(Diagrams are schematic and not to scale)

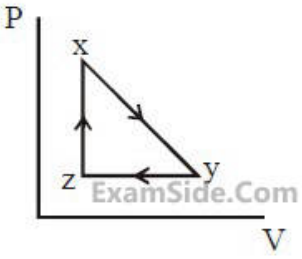
(A)



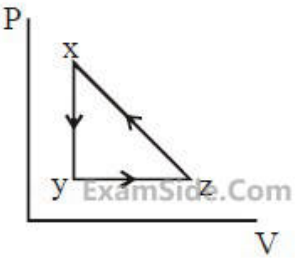
(B)



(C)



(D)



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Q102. JEE Main 2020 (Online) 7th January Morning Slot

A litre of dry air at STP expands adiabatically to a volume of 3 litres. If γ = 1.40, the work done by air is : ($3^{1.4}$ = 4.6555) [Take air to be an ideal gas]

(A) 60.7 J (B) 100.8 J (C) 90.5 J (D) 48 J

Open solution

Q103. JEE Main 2019 (Online) 12th April Evening Slot

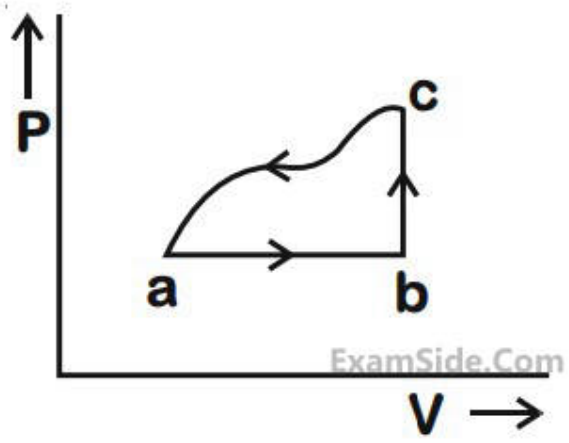
A diatomic gas with rigid molecules does 10 J of work when expanded at constant pressure. What would be the heat energy absorbed by the gas, in this process ?

(A) 35 J (B) 30 J (C) 25 J (D) 40 J

Open solution

Q104. JEE Main 2019 (Online) 12th April Morning Slot

A sample of an ideal gas is taken through the cyclic process abca as shown in the figure. The change in the internal energy of the gas along the path ca is −180 J. The gas absorbs 250 J of heat along the path ab and 60 J along the path bc. The work done by the gas along the path abc is:



(A) 120 J (B) 130 J (C) 100 J (D) 140 J

Open solution

Q105. JEE Main 2019 (Online) 10th April Evening Slot

One mole of ideal gas passes through a process where pressure and volume obey the relation $P = P_0\left[1 - \frac{1}{2}\left(\frac{V_0}{V}\right)^2\right]$. Here P_0 and V_0 are constants. Calculate the change in the temperature of the gas if its volume changes form V_0 to $2V_0$

(A) $\frac{3}{4}\frac{P_0V_0}{R}$ (B) $\frac{1}{2}\frac{P_0V_0}{R}$

(C) $\frac{5}{4}\frac{P_0V_0}{R}$ (D) $\frac{1}{4}\frac{P_0V_0}{R}$

Open solution

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Q106. JEE Main 2019 (Online) 10th April Evening Slot

When heat Q is supplied to a diatomic gas of rigid molecules, at constant volume its temperature increases by ΔT. the heat required to produce the same change in temperature, at a constant pressure is :

Open solution

(A) $\frac{7}{5}Q$ (B) $\frac{3}{2}Q$ (C) $\frac{2}{3}Q$ (D) $\frac{5}{3}Q$

Q107. JEE Main 2019 (Online) 10th April Morning Slot

A cylinder with fixed capacity of 67.2 lit contains helium gas at STP. The amount of heat needed to raise the temperature of the gas by 20°C is : [Given that R = 8.31 J mol^{-1} K^{-1}]

Open solution

(A) 374 J (B) 700 J (C) 748 J (D) 350 J

Q108. JEE Main 2019 (Online) 10th April Morning Slot

n moles of an ideal gas with constant volume heat capcity C_V undergo an isobaric expansion by certain volume. The ratio of the work done in the process, to the heat supplied is :

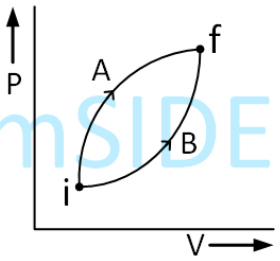
Open solution

(A) $\frac{nR}{C_V-nR}$ (B) $\frac{4nR}{C_V-nR}$ (C) $\frac{4nR}{C_V+nR}$ (D) $\frac{nR}{C_V+nR}$

Q109. JEE Main 2019 (Online) 9th April Morning Slot

Following figure shows two processes A and B for a gas. If ΔQ_A and ΔQ_B are the amount of heat absorbed by the system in two cases, and ΔU_A and ΔU_B are changes in internal energies, respectively, then :

Open solution



ExamSIDE.com

(A) $\Delta Q_A > \Delta Q_B$; $\Delta U_A > \Delta U_B$ (B) $\Delta Q_A < \Delta Q_B$; $\Delta U_A < \Delta U_B$

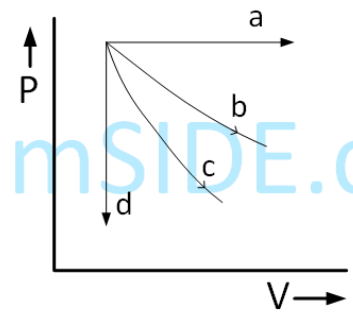
(C) $\Delta Q_A > \Delta Q_B$; $\Delta U_A = \Delta U_B$ (D) $\Delta Q_A = \Delta Q_B$; $\Delta U_A = \Delta U_B$

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Q110. JEE Main 2019 (Online) 8th April Evening Slot

The given diagram shows four processes i.e., isochoric, isobaric, isothermal and adiabatic. The correct assignment of the processes, in the same order is given by :-

Open solution



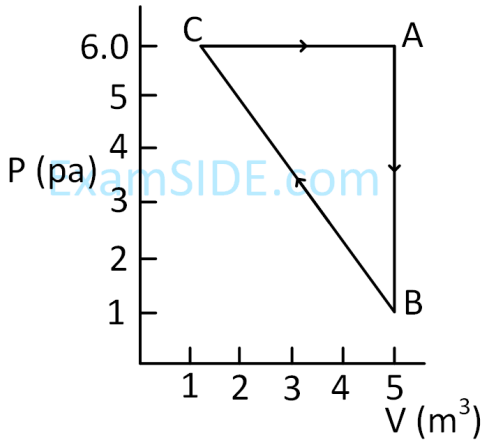
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(A) d a b c (B) a d b c (C) d a c b (D) a d c b

Q111. JEE Main 2019 (Online) 12th January Morning Slot

For the given cyclic process CAB as shown for a gas, the work done is :

Open solution



(A) 1 J (B) 10 J (C) 5 J (D) 30 J

Q112. JEE Main 2019 (Online) 11th January Evening Slot

In a process, temperature and volume of one mole of an ideal monoatomic gas are varied according to the relation VT = K, where K is a constant. In this process the temperature of the gas is increased by ΔT. The amount of heat absorbed by gas is (R is gas constant) :

Open solution

(A) $\frac{1}{2}$ KRΔT (B) $\frac{1}{2}$ RΔT (C) $\frac{3}{2}$ RΔT (D) $\frac{2K}{3}$ ΔT

Q113. JEE Main 2019 (Online) 11th January Morning Slot

A rigid diatomic ideal gas undergoes an adiabatic process at room temperature. The relation between temperature and volume for this process is TV^x = constant, then x is :

Open solution

(A) $\frac{5}{3}$ (B) $\frac{2}{5}$ (C) $\frac{3}{5}$ (D) $\frac{2}{3}$

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Q114. JEE Main 2019 (Online) 10th January Evening Slot

Open solution

Half mole of an ideal monoatomic gas is heated at constant pressure of 1 atm from 20°C to 90°C. Work done by gas is close to – (Gas constant R = 8.31 J/mol.K)

(A) 581 J (B) 73 J (C) 146 J (D) 291 J

Q115. JEE Main 2019 (Online) 9th January Evening Slot

Open solution

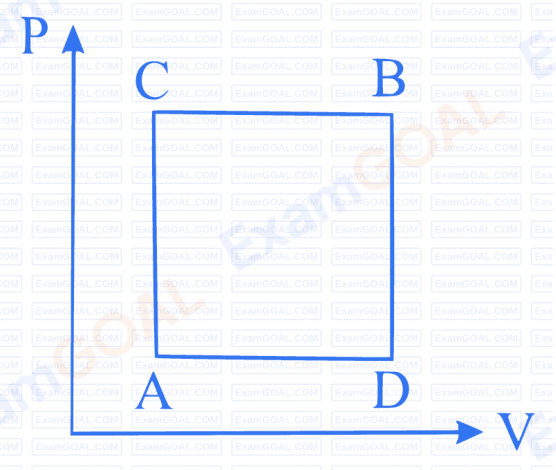
A 15 g mass of nitrogen gas is enclosed in a vessel at a temperature 27°C. Amount of heat transferred to the gas, so that rms velocity of molecules is doubled, is about : [Take R = 8.3 J/K mole]

(A) 0.9 kJ (B) 6 kJ (C) 10 kJ (D) 14 kJ

Q116. JEE Main 2019 (Online) 9th January Morning Slot

Open solution

A gas can be taken from A to B via two different processes ACB and ADB.



When path ACB is used 60 J of heat flows into the system and 30 J of work is done by the system. If path ADB is used work done by the system is 10 J. The heat Flow into the system in path ADB is :

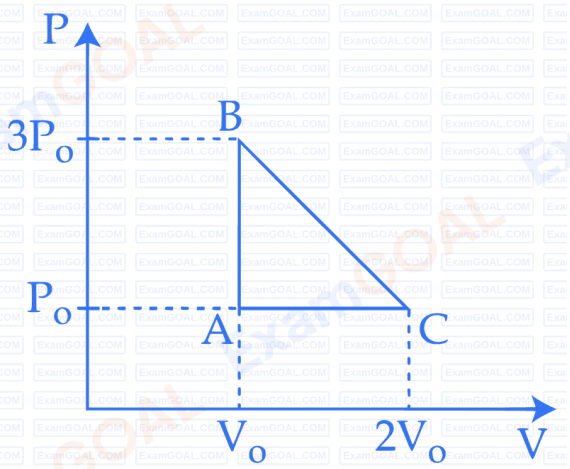
(A) 40 J (B) 80 J (C) 100 J (D) 20 J

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Q117. JEE Main 2018 (Online) 16th April Morning Slot

One mole of an ideal monoatomic gas is taken along the path ABCA as shown in the PV diagram. The maximum temperature attained by the gas along the path BC is given by :

Open solution



(A) $\frac{25}{16}\frac{P_oV_o}{R}$

(B) $\frac{25}{8}\frac{P_oV_o}{R}$

(C) $\frac{25}{4}\frac{P_oV_o}{R}$

(D) $\frac{5}{8}\frac{P_oV_o}{R}$

Q118. JEE Main 2018 (Offline)

Two moles of an ideal monatomic gas occupies a volume V at 27°C. The gas expands adiabatically to a volume 2 V. Calculate (a) the final temperature of the gas and (b) change in its internal energy.

Open solution

(A) (a) 195 K (b) 2.7 kJ

(B) (a) 189 K (b) 2.7 kJ

(C) (a) 195 K (b) −2.7 kJ

(D) (a) 189 K (b) − 2.7 kJ

Q119. JEE Main 2018 (Online) 15th April Morning Slot

One mole of an ideal monoatomic gas is compressed isothermally in a rigid vessel to double its pressure at room temperature, $27^{\circ}C$. The work done on the gas will be :

Open solution

(A) $300\ R$

(B) $300\ R\ ln\ 6$

(C) $300\ R\ ln\ 2$

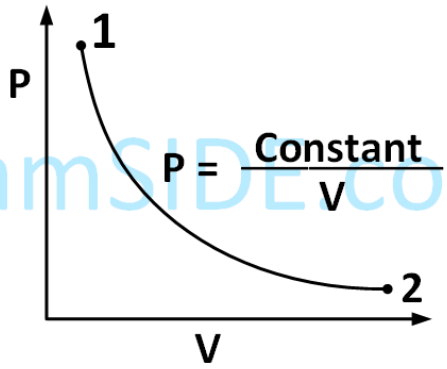
(D) $300\ R\ ln\ 7$

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Q120. JEE Main 2017 (Online) 9th April Morning Slot

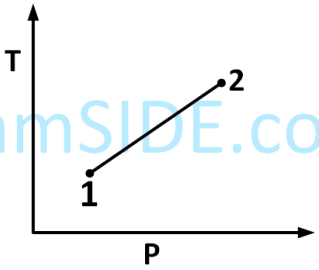
Open solution

For the P-V diagram given for an ideal gas,

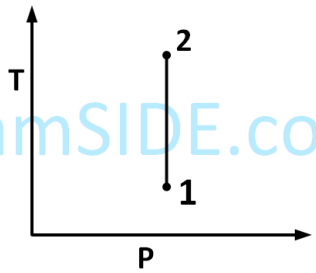


out of the following which one correctly represents the T-P diagram ?

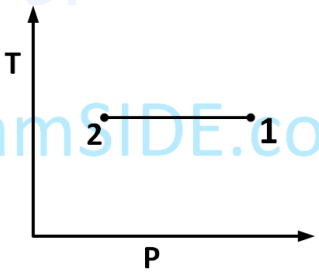
(A)



(B)

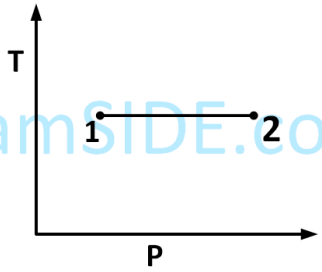


(C)



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(D)



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Q121. JEE Main 2017 (Offline)

Open solution

C_P and C_V are specific heats at constant pressure and constant volume respectively. It is observed that

$C_P - C_V$ = a for hydrogen gas

$C_P - C_V$ = b for nitrogen gas

The correct relation between a and b is

(A) a = 28 b (B) a = 1/14 b (C) a = b (D) a = 14 b

Q122. JEE Main 2016 (Online) 9th April Morning Slot

Open solution

The ratio of work done by an ideal monoatomic gas to the heat supplied to it in an isobaric process is :

(A) $\frac{3}{5}$ (B) $\frac{2}{3}$ (C) $\frac{3}{2}$ (D) $\frac{2}{5}$

Q123. JEE Main 2016 (Offline)

Open solution

An ideal gas undergoes a quasi static, reversible process in which its molar heat capacity C remains constant. If during this process the relation of pressure P and volume V is given by $PV^n =$ constant, then n is given by (Here C_p and C_v are molar specific heat at constant pressure and constant volume, respectively:

(A) $n = \frac{C_p - C}{C - C_v}$ (B) $n = \frac{C - C_v}{C - C_p}$

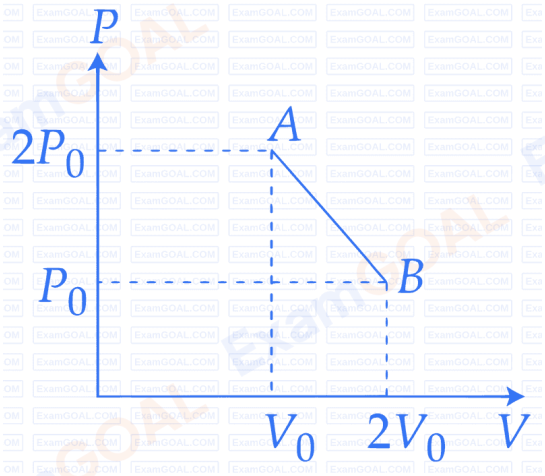
(C) $n = \frac{C_p}{C_v}$ (D) $n = \frac{C - C_p}{C - C_v}$

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Q124. JEE Main 2016 (Offline)

Open solution

$'n'$ moles of an ideal gas undergoes a process $A \rightarrow B$ as shown in the figure. The maximum temperature of the gas during the process will be :



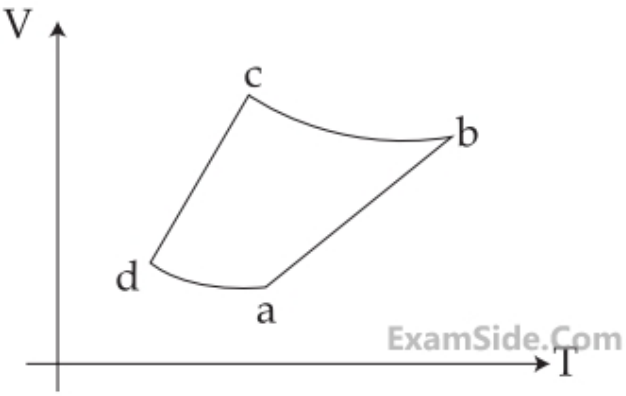
(A) $\frac{9P_0V_0}{2nR}$ (B) $\frac{9P_0V_0}{nR}$ (C) $\frac{9P_0V_0}{4nR}$ (D) $\frac{3P_0V_0}{2nR}$

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Q125. JEE Main 2015 (Online) 10th April Morning Slot

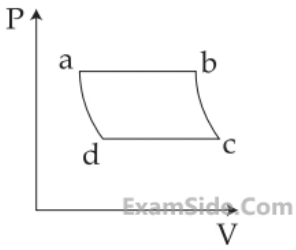
Open solution

An ideal gas goes through a reversible cycle a → b → c → d which has the V − T diagram shown below. Process d → a and b → c are adiabatic.

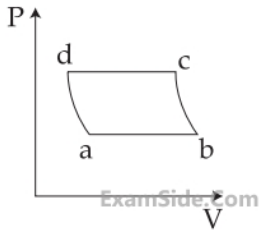


The corresponding P - V diagram for the process is (all figures are schematic and not drawn to scale) :

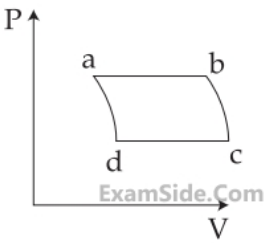
(A)



(B)

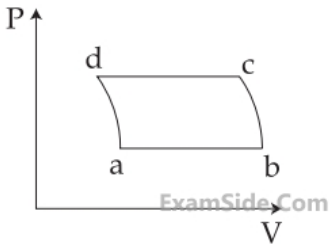


(C)



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(D)



Q126. JEE Main 2015 (Offline)

A solid body of constant heat capacity $1\ J/^{\circ}C$ is being heated by keeping it in contact with reservoirs in two ways:

(i) Sequentially keeping in contact with 2 reservoirs such that each reservoir supplies same amount of heat.

(ii) Sequentially keeping in contact with 8 reservoirs such that each reservoir supplies same amount of heat.

In both the cases body is brought from initial temperature $100^{\circ}C$ to final temperature $200^{\circ}C$. Entropy change of the body in the two cases respectively is :

(A) $ln2, 2ln2$ (B) $2ln2, 8ln2$ (C) $ln2, 4ln2$ (D) $ln2, ln2$

Open solution

Q127. JEE Main 2015 (Offline)

Consider a spherical shell of radius R at temperature T. The black body radiation inside it can be considered as an ideal gas of photons with internal energy per unit volume $u = \frac{U}{V} \propto T^4$ and pressure $p = \frac{1}{3}\left(\frac{U}{V}\right)$. If the shell now undergoes an adiabatic expansion the relation between T and R is:

(A) $T \propto \frac{1}{R}$ (B) $T \propto \frac{1}{R^3}$ (C) $T \propto e^{-R}$ (D) $T \propto e^{-3R}$

Open solution

Q128. JEE Main 2014 (Online) 19th April Morning Slot

A gas is compressed from a volume of 2 m^3 to a volume of 1 m^3 at a constant pressure of 100 N/m^2. Then it is heated at constant volume by supplying 150 J of energy. As a result, the internal energy of the gas :

(A) Increases by 250 J (B) Decreases by 250 J

(C) Increases by 50 J (D) Decreases by 50 J

Open solution

Q129. JEE Main 2014 (Online) 11th April Morning Slot

During an adiabatic compression, 830 J of work is done on 2 moles of a diatomic ideal gas to reduce its volume by 50%. The change in its temperature is nearly : (R = 8.3 JK^{-1} mol^{-1})

(A) 40 K (B) 33 K (C) 20 K (D) 14 K

Open solution

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Q130. JEE Main 2014 (Online) 9th April Morning Slot

Open solution

The equation of state for a gas is given by PV = nRT + αV, where n is the number of moles and α is a positive constant. The initial temperature and pressure of one mole of the gas contained in a cylinder are T_o and P_o respectively. The work done by the gas when its temperature doubles isobarically will be :

(A) $\frac{P_o T_o R}{P_o - \alpha}$

(B) $\frac{P_o T_o R}{P_o + \alpha}$

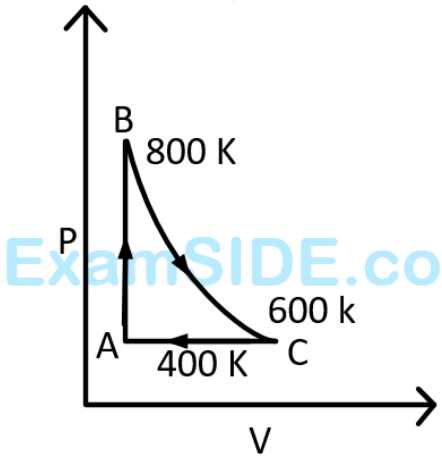
(C) $P_o T_o R \ln 2$

(D) $P_o T_o R$

Q131. JEE Main 2014 (Offline)

Open solution

One mole of a diatomic ideal gas undergoes a cyclic process ABC as shown in figure. The process BC is adiabatic. The temperatures at A, B and C are $400\ K$, $800\ K$ and $600\ K$ respectively. Choose the correct statement :



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(A) The change in internal energy in whole cyclic process is $250\ R$.

(B) The change in internal energy in the process CA is $700\ R$.

(C) The change in internal energy in the process AB is - $350\ R$.

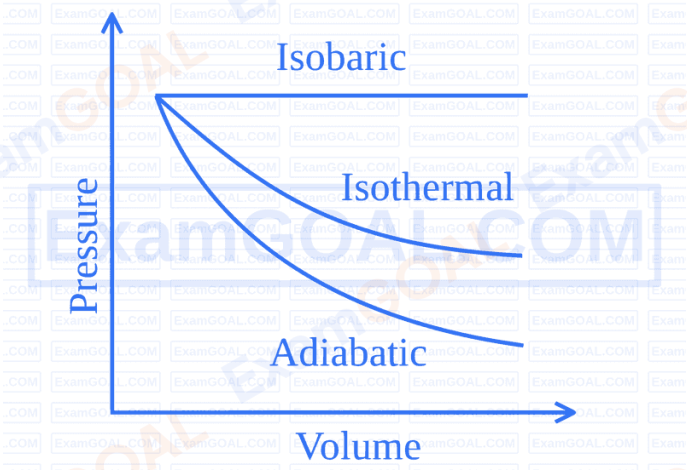
(D) The change in internal energy in the process BC is - $500\ R$.

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Q132. JEE Main 2013 (Online) 25th April Morning Slot

A sample of gas expands from V_1 to V_2. In which of the following, the work done will be greatest?

Open solution



(A) Same in all processes
(B) Isobaric process
(C) Isothermal process
(D) Adiabatic process

Q133. JEE Main 2013 (Online) 23th April Morning Slot

This question has Statement-1 and Statement-2. Of the four choices given after the Statements, choose the one that best describes the two Statements.
Statement 1: The internal energy of a perfect gas is entirely kinetic and depends only on absolute temperature of the gas and not on its pressure or volume.
Statement 2: A perfect gas is heated keeping pressure constant and later at constant volume. For the same amount of heat the temperature of the gas at constant pressure is lower than that at constant volume.

Open solution

(A) Statement-1 is true, Statement-2 is true and Statement-2 is the correct explanation of Statement-1.
(B) Statement-1 is true, Statement-2 is false.
(C) Statement-1 is true, Statement-2 is true but Statement-2 is not the correct explanation of Statement-1.
(D) Statement-1 is false, Statement-2 is true.

Q134. JEE Main 2013 (Online) 22th April Morning Slot

Given that 1 g of water in liquid phase has volume 1 cm^3 and in vapour phase 1671 cm^3 at atmospheric pressure and the latent heat of vaporization of water is 2256 J/g; the change in the internal energy in joules for 1 g of water at 373 K when it changes from liquid phase to vapour phase at the same temperature is :

Open solution

(A) 2256 (B) 167 (C) 2089 (D) 1

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Q135. JEE Main 2013 (Online) 22th April Morning Slot

Open solution

An ideal gas at atmospheric pressure is adiabatically compressed so that its density becomes 32 times of its initial value. If the final pressure of gas is 128 atmospheres, the value of 'γ' of the gas is :

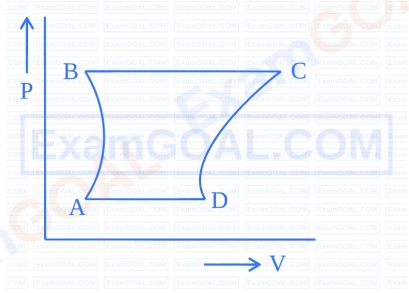
(A) 1.5 (B) 1.4 (C) 1.3 (D) 1.6

Q136. JEE Main 2013 (Online) 22th April Morning Slot

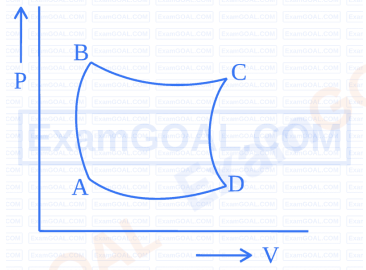
Open solution

A certain amount of gas is taken through a cyclic process (A B C D A) that has two isobars, one isochore and one isothermal. The cycle can be represented on a P - V indicator diagram as :

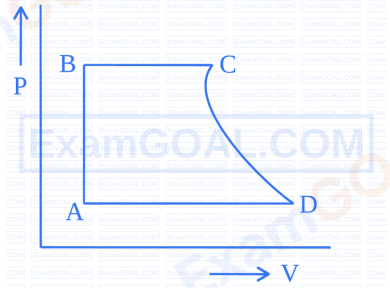
(A)



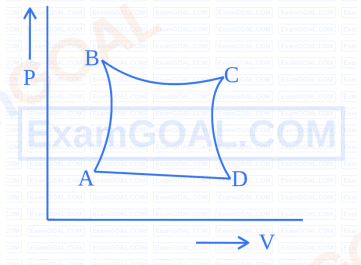
(B)



(C)



(D)

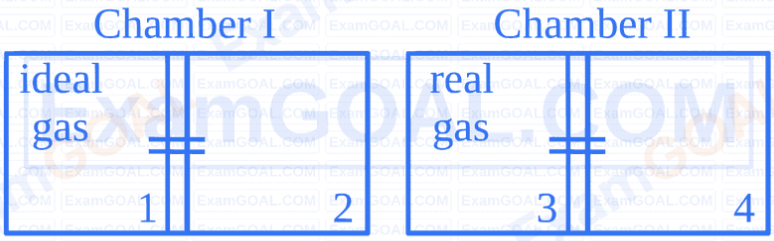


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Q137. JEE Main 2013 (Online) 9th April Morning Slot

Open solution

There are two identical chambers, completely thermally insulated from surroundings. Both chambers have a partition wall dividing the chambers in two compartments. Compartment 1 is filled with an ideal gas and Compartment 3 is filled with a real gas. Compartments 2 and 4 are vacuum. A small hole (orifice) is made in the partition walls and the gases are allowed to expand in vacuum.
Statement - 1 : No change in the temperature of the gas takes place when ideal gas expands in vacuum. However, the temperature of real gas goes down (cooling) when it expands in vacuum.
Statement - 2 : The internal energy of an ideal gas is only kinetic. The internal energy of a real gas is kinetic as well as potential.



(A) Statement-1 is false and Statement-2 is true.

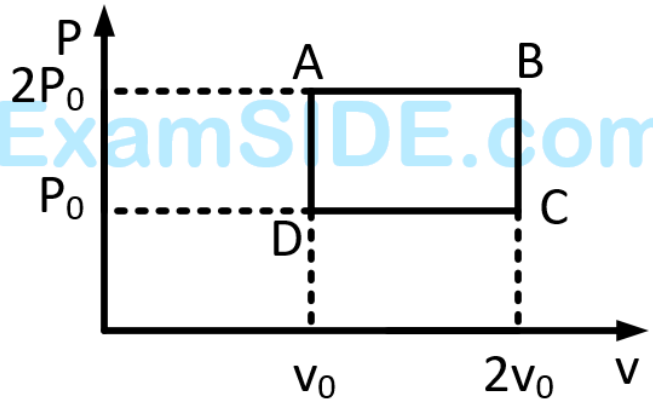
(B) Statement-1 and Statement-2 both are true. Statement-2 is the correct explanation of Statement-1.

(C) Statement-1 is true and Statement-2 is false.

(D) Statement-1 and Statement-2 both are true. Statement-2 is not correct explanation of Statement-1.

Q138. JEE Main 2013 (Offline)

Open solution



The above p-v diagram represents the thermodynamic cycle of an engine, operating with an ideal monatomic gas. The amount of heat, extracted from the source in a single cycle is

(A) p_0v_0

(B) $\left(\frac{13}{2}\right)p_0v_0$

(C) $\left(\frac{11}{2}\right)p_0v_0$

(D) $4p_0v_0$

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Q139. AIEEE 2012 (Online) 26th May

An ideal monatomic gas with pressure P, volume V and temperature T is expanded isothermally to a volume $2V$ and a final pressure P_i. If the same gas is expanded adiabatically to a volume $2V$, the final pressure is P_a. The ratio $\frac{P_a}{P_i}$ is

Open solution

(A) $2^{-1/3}$ (B) $2^{1/3}$ (C) $2^{2/3}$ (D) $2^{-2/3}$

Q140. AIEEE 2012 (Online) 19th May

The pressure of an ideal gas varies with volume as $P = \alpha V$, where α is a constant. One mole of the gas is allowed to undergo expansion such that its volume becomes 'm' times its initial volume. The work done by the gas in the process is

Open solution

(A) $\frac{\alpha V}{2}(m^2 - 1)$ (B) $\frac{\alpha^2 V^2}{2}(m^2 - 1)$
(C) $\frac{\alpha}{2}(m^2 - 1)$ (D) $\frac{\alpha V^2}{2}(m^2 - 1)$

Q141. AIEEE 2012 (Online) 12th May

A given ideal gas with $\gamma = \frac{C_p}{C_v} = 1.5$ at a temperature T. If the gas is compressed adiabatically to one-fourth of its initial volume, the final temperature will be

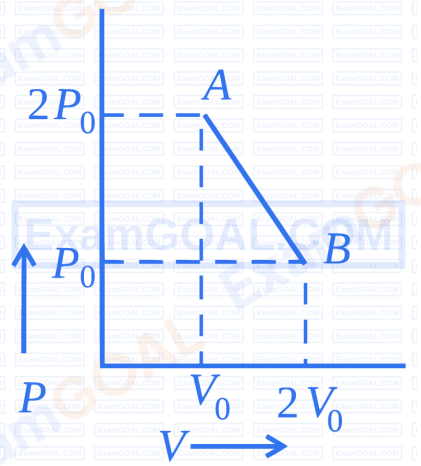
Open solution

(A) $2\sqrt{2}T$ (B) $4T$ (C) $2T$ (D) $8T$

Q142. AIEEE 2012 (Online) 12th May

n moles of an ideal gas undergo a process $A \rightarrow B$ as shown in the figure. Maximum temperature of the gas during the process is

Open solution



(A) $\frac{9P_0V_0}{nR}$ (B) $\frac{3P_0V_0}{2nR}$ (C) $\frac{9P_0V_0}{2nR}$ (D) $\frac{9P_0V_0}{4nR}$

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Q143. AIEEE 2012 (Online) 7th May

Open solution

This question has Statement 1 and Statement 2. Of the four choices given after the Statements, choose the one that best describes the two Statements.
Statement 1: In an adiabatic process, change in internal energy of a gas is equal to work done on/by the gas in the process.
Statement 2: The temperature of a gas remains constant in an adiabatic process.

(A) Statement 1 is true, Statement 2 is true, Statement 2 is a correct explanation of Statement 1.

(B) Statement 1 is true, Statement 2 is false.

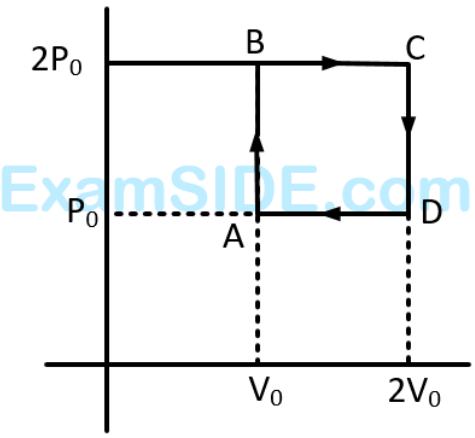
(C) Statement 1 is false, Statement 2 is true.

(D) Statement 1 is false, Statement 2 is true, Statement 2 is not a correct explanation of Statement 1.

Q144. AIEEE 2012

Open solution

Helium gas goes through a cycle $ABCD$ (consisting of two isochoric and isobaric lines) as shown in figure efficiency of this cycle is nearly : (Assume the gas to be close to ideal gas)



(A) 15.4% (B) 9.1% (C) 10.5% (D) 12.5%

Q145. AIEEE 2011

Open solution

A thermally insulated vessel contains an ideal gas of molecular mass M and ratio of specific heats γ. It is moving with speed v and it's suddenly brought to rest. Assuming no heat is lost to the surroundings, Its temperature increases by:

(A) $\frac{(\gamma-1)}{2\gamma R}Mv^2K$ (B) $\frac{\gamma M^2 v}{2R}K$

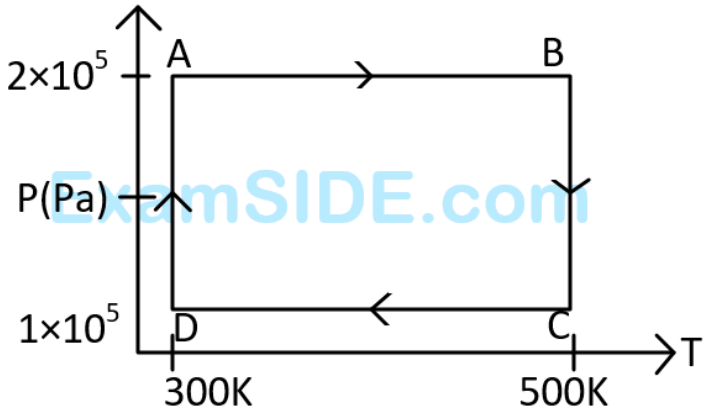
(C) $\frac{(\gamma-1)}{2R}Mv^2K$ (D) $\frac{(\gamma-1)}{2(\gamma+1)R}Mv^2K$

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Q146. AIEEE 2009

Two moles of helium gas are taken over the cycle $ABCD$, as shown in the P-T diagram.

Open solution



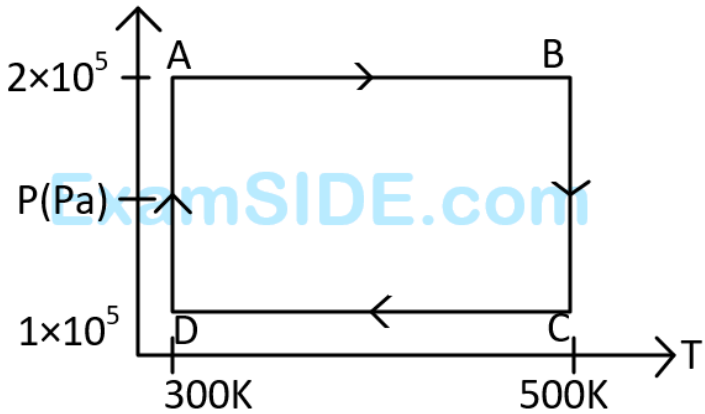
Assuming the gas to be ideal the work done on the gas in taking it from A to B is :

(A) $300\ R$ (B) $400\ R$ (C) $500\ R$ (D) $200\ R$

Q147. AIEEE 2009

Two moles of helium gas are taken over the cycle $ABCD$, as shown in the P-T diagram.

Open solution



The work done on the gas in taking it from D to A is :

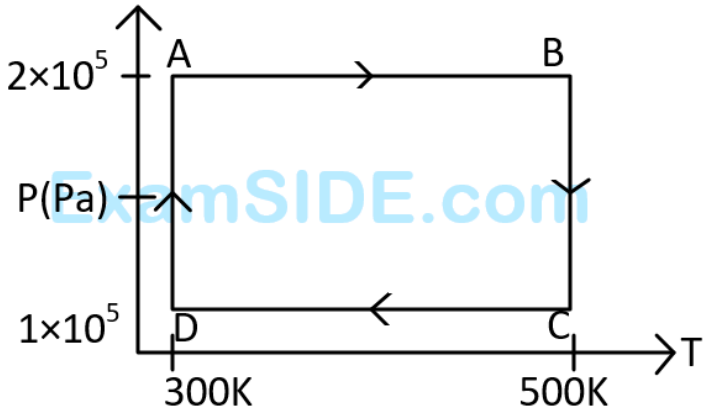
(A) $+414\ R$ (B) $-690\ R$ (C) $+690\ R$ (D) $-414\ R$

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Q148. AIEEE 2009

Open solution

Two moles of helium gas are taken over the cycle $ABCD$, as shown in the P-T diagram.



ExamSIDE.com

The net work done on the gas in the cycle $ABCDA$ is:

(A) $276\ R$ (B) $1076\ R$ (C) $1904\ R$ (D) zero

Q149. AIEEE 2008

Open solution

An insulated container of gas has two chambers separated by an insulating partition. One of the chambers has volume V_1 and contains ideal gas at pressure P_1 and temperature T_1. The other chamber has volume V_2 and contains ideal gas at pressure P_2 and temperature T_2. If the partition is removed without doing any work on the gas, the final equilibrium temperature of the gas in the container will be

(A) $\frac{T_1T_2(P_1V_1+P_2V_2)}{P_1V_1T_2+P_2V_2T_1}$

(B) $\frac{P_1V_1T_1+P_2V_2T_2}{P_1V_1+P_2V_2}$

(C) $\frac{P_1V_1T_2+P_2V_2T_1}{P_1V_1+P_2V_2}$

(D) $\frac{T_1T_2(P_1V_1+P_2V_2)}{P_1V_1T_1+P_2V_2T_2}$

Q150. AIEEE 2007

Open solution

If C_p and C_v denote the specific heats of nitrogen per unit mass at constant pressure and constant volume respectively, then

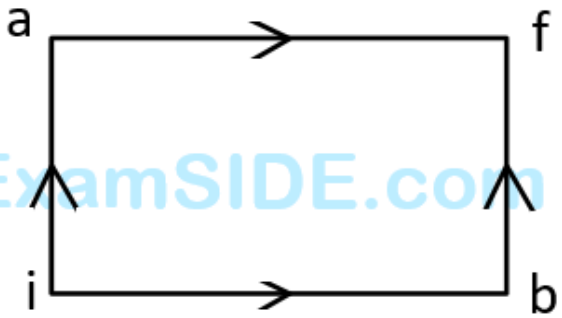
(A) $C_p - C_v = 28R$ (B) $C_p - C_v = R/28$ (C) $C_p - C_v = R/14$ (D) $C_p - C_v = R$

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Q151. AIEEE 2007

When a system is taken from state i to state f along the path iaf, it is found that $Q = 50$ cal and $W = 20\ cal$. Along the path $ibf\ Q = 36\ cal$. W along the path ibf is

Open solution



(A) $14\ cal$ (B) $6\ cal$ (C) $16\ cal$ (D) $66\ cal$

Q152. AIEEE 2006

The work of $146\ kJ$ is performed in order to compress one kilo mole of gas adiabatically and in this process the temperature of the gas increases by $7^{\circ}C$. The gas is $\left(R = 8.3J\ mol^{-1}\ K^{-1}\right)$

Open solution

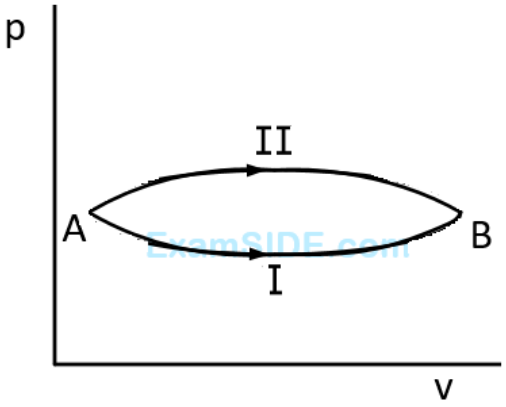
(A) diatomic (B) triatomic

(C) a mixture of monoatomic and diatomic (D) monoatomic

Q153. AIEEE 2005

A system goes from A to B via two processes I and II as shown in figure. If ΔU_1 and ΔU_2 are the changes in internal energies in the processes I and II respectively, then

Open solution



(A) relation between ΔU_1 and ΔU_2 can not be determined

(B) $\Delta U_1 = \Delta U_2$

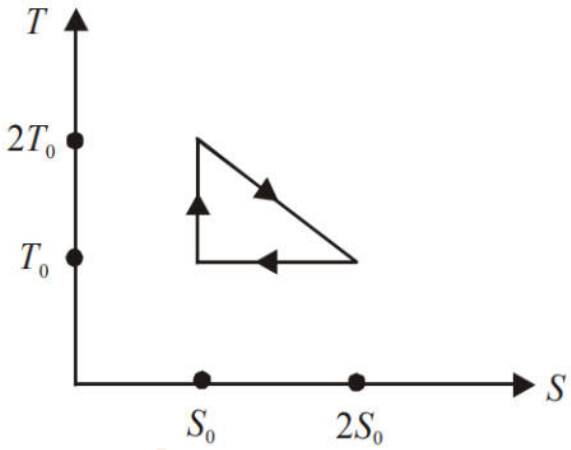
(C) $\Delta U_2 < \Delta U_1$

(D) $\Delta U_2 > \Delta U_1$

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Q154. AIEEE 2005

The temperature-entropy diagram of a reversible engine cycle is given in the figure. Its efficiency is



Open solution

(A) $\frac{1}{4}$ (B) $\frac{1}{2}$ (C) $\frac{2}{3}$ (D) $\frac{1}{3}$

Q155. AIEEE 2004

Which of the following statements is **correct** for any thermodynamic system ?

(A) The change in entropy can never be zero

(B) Internal energy and entropy are state functions

(C) The internal energy changes in all processes

(D) The work done in an adiabatic process is always zero

Open solution

Q156. AIEEE 2004

Two thermally insulated vessels 1 and 2 are filled with air at temperatures (T_1, T_2), volume (V_1, V_2) and pressure (P_1, P_2) respectively. If the valve joining the two vessels is opened, the temperature inside the vessel at equilibrium will be

(A) $T_1T_2\,(P_1V_1 + P_2V_2)/\,(P_1V_1T_2 + P_2V_2T_1)$

(B) $(T_1 + T_2)/2$

(C) $T_1 + T_2$

(D) $T_1T_2\,(P_1V_1 + P_2V_2)/\,(P_1V_1T_1 + P_2V_2T_2)$

Open solution

Q157. AIEEE 2003

"Heat cannot by itself flow from a body at lower temperature to a body at higher temperature" is a statement or consequence of :

(A) second law of thermodynamics (B) conservation of momentum

(C) conservation of mass (D) first law of thermodynamics

Open solution

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Q158. AIEEE 2003

During an adiabatic process, the pressure of a gas is found to be proportional to the cube of its absolute temperature. The ratio C_p/C_V for the gas is

(A) $\frac{4}{3}$ (B) 2 (C) $\frac{5}{3}$ (D) $\frac{3}{2}$

Open solution

Q159. AIEEE 2003

Which of the following parameters does not characterize the thermodynamic state of mattter?

(A) Temperature (B) Pressure (C) Work (D) Volume

Open solution

Numerical 24 questions

Q160. JEE Main 2026 (Online) 2nd April Evening Shift

5 moles of unknown gas is heated at constant volume from 10°C to 20°C. The molar specific heat of this gas at constant pressure $c_p = 8$ cal/mol.°C and $R = 8.36$ J/mol.°C. The change in the internal energy of the gas is _____ calorie.

Open solution

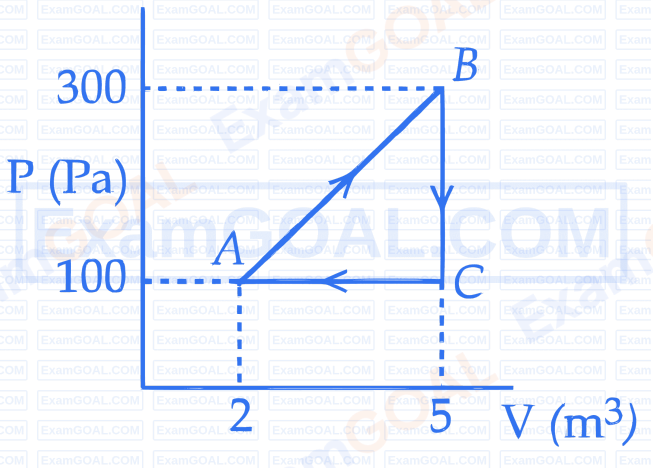
Q161. JEE Main 2026 (Online) 2nd April Morning Shift

A vessel contains 0.15 m^3 of a gas at pressure 8 bar and temperature 140 °C with $c_p = 3R$ and $c_v = 2R$. It is expanded adiabatically till pressure falls to 1 bar. The work done during this process is _____ kJ. (R is gas constant)

Open solution

Q162. JEE Main 2026 (Online) 28th January Evening Shift

A thermodynamic system is taken through the cyclic process *ABC* as shown in the figure. The total work done by the system during the cycle *ABC* is _____ J.



Open solution

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Q163. JEE Main 2026 (Online) 24th January Evening Shift

When 300 J of heat given to an ideal gas with $C_p = \frac{7}{2}R$ its temperature raises from 20°C to 50°C keeping its volume constant. The mass of the gas is (approximately) ____ g. $(\text{R} = 8.314\,\text{J/mol. K})$

Open solution

Q164. JEE Main 2026 (Online) 21st January Evening Shift

A diatomic gas $(\gamma = 1.4)$ does 100 J of work when it is expanded isobarically. Then the heat given to the gas ____ J.

Open solution

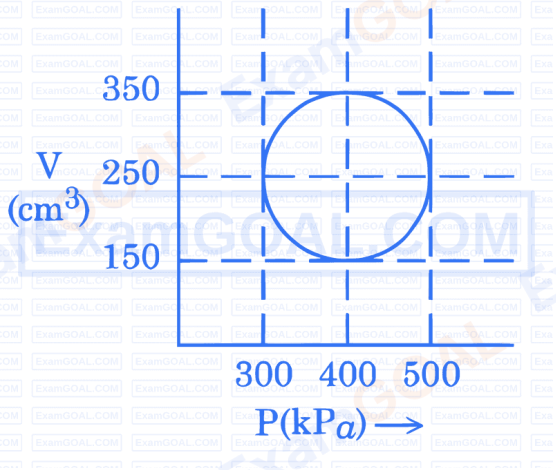
Q165. JEE Main 2026 (Online) 21st January Morning Shift

10 mole of oxygen is heated at constant volume from 30°C to 40°C. The change in the internal energy of the gas is ____ cal. (The molecular specific heat of oxygen at constant pressure, $C_P = 7\text{cal/mol.}^\circ\text{C}$ and $\text{R} = 2\text{cal.}/\text{mol.}^\circ\text{C.}$)

Open solution

Q166. JEE Main 2025 (Online) 7th April Morning Shift

An ideal gas has undergone through the cyclic process as shown in the figure. Work done by the gas in the entire cycle is _____ $\times 10^{-1}$ J. (Take $\pi = 3.14$)



Open solution

Q167. JEE Main 2025 (Online) 24th January Morning Shift

The temperature of 1 mole of an ideal monoatomic gas is increased by 50°C at constant pressure. The total heat added and change in internal energy are E_1 and E_2, respectively. If $\frac{E_1}{E_2} = \frac{x}{9}$ then the value of x is _____.

Open solution

Q168. JEE Main 2025 (Online) 23rd January Morning Shift

An ideal gas initially at 0°C temperature, is compressed suddenly to one fourth of its volume. If the ratio of specific heat at constant pressure to that at constant volume is $3/2$, the change in temperature due to the thermodynamic process is _____ K.

Open solution

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Q169. JEE Main 2022 (Online) 29th July Morning Shift

The pressure P_1 and density d_1 of diatomic gas $\left(\gamma = \frac{7}{5}\right)$ changes suddenly to $P_2\,(> P_1)$ and d_2 respectively during an adiabatic process. The temperature of the gas increases and becomes _____ times of its initial temperature. (given $\frac{d_2}{d_1} = 32$)

Open solution

Q170. JEE Main 2022 (Online) 28th July Evening Shift

At a certain temperature, the degrees of freedom per molecule for gas is 8. The gas performs 150 J of work when it expands under constant pressure. The amount of heat absorbed by the gas will be _____ J.

Open solution

Q171. JEE Main 2022 (Online) 27th June Evening Shift

A diatomic gas (γ = 1.4) does 400J of work when it is expanded isobarically. The heat given to the gas in the process is _____ J.

Open solution

Q172. JEE Main 2022 (Online) 24th June Evening Shift

A monoatomic gas performs a work of $\frac{Q}{4}$ where Q is the heat supplied to it. The molar heat capacity of the gas will be _____ R during this transformation. Where R is the gas constant.

Open solution

Q173. JEE Main 2022 (Online) 24th June Morning Shift

0.056 kg of Nitrogen is enclosed in a vessel at a temperature of 127°C. The amount of heat required to double the speed of its molecules is _____ k cal.

(Take R = 2 cal mole^{-1} K^{-1})

Open solution

Q174. JEE Main 2021 (Online) 1st September Evening Shift

The temperature of 3.00 mol of an ideal diatomic gas is increased by 40.0°C without changing the pressure of the gas. The molecules in the gas rotate but do not oscillate. If the ratio of change in internal energy of the gas to the amount of workdone by the gas is $\frac{x}{10}$. Then the value of x (round off to the nearest integer) is _____. (Given R = 8.31 J mol^{-1} K^{-1})

Open solution

Q175. JEE Main 2021 (Online) 31st August Evening Shift

A sample of gas with γ = 1.5 is taken through an adiabatic process in which the volume is compressed from 1200 cm^3 to 300 cm^3. If the initial pressure is 200 kPa. The absolute value of the workdone by the gas in the process = _____ J.

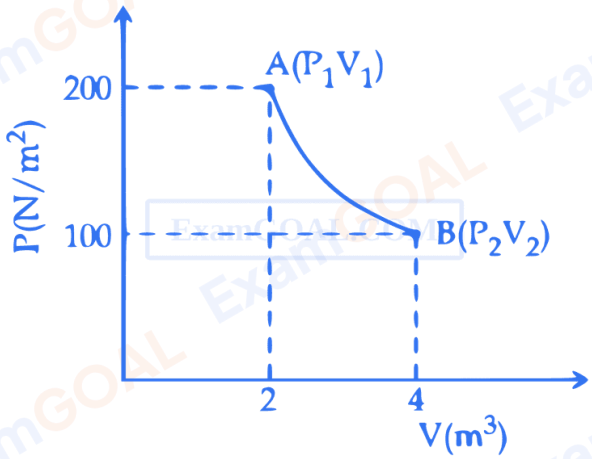
Open solution

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Q176. JEE Main 2021 (Online) 20th July Evening Shift

One mole of an ideal gas at 27°C is taken from A to B as shown in the given PV indicator diagram. The work done by the system will be _____ $\times 10^{-1}$ J. [Given : R = 8.3 J/mole K, ln2 = 0.6931] (Round off to the nearest integer)

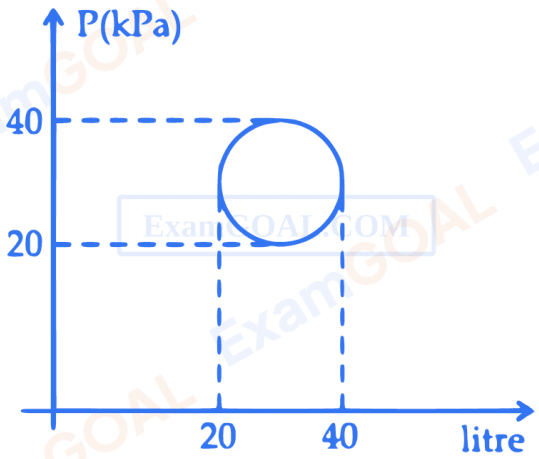
Open solution



Q177. JEE Main 2021 (Online) 20th July Morning Shift

In the reported figure, heat energy absorbed by a system in going through a cyclic process is _____ πJ.

Open solution



Q178. JEE Main 2021 (Online) 26th February Evening Shift

1 mole of rigid diatomic gas performs a work of $\frac{Q}{5}$ when heat Q is supplied to it. The molar heat capacity of the gas during this transformation is $\frac{xR}{8}$. The value of x is _____. [R = universal gas constant]

Open solution

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Q179. JEE Main 2021 (Online) 26th February Evening Shift

The volume V of a given mass of monoatomic gas changes with temperature T according to the relation $V = KT^{\frac{2}{3}}$. The workdone when temperature changes by 90K will be xR. The value of x is _____. [R = universal gas constant]

Open solution

Q180. JEE Main 2021 (Online) 25th February Morning Shift

A monoatomic gas of mass 4.0 u is kept in an insulated container. Container is moving with velocity 30 m/s. If container is suddenly stopped then change in temperature of the gas (R = gas constant) is $\frac{x}{3R}$. Value of x is _____.

Open solution

Q181. JEE Main 2021 (Online) 25th February Morning Shift

In a certain thermodynamical process, the pressure of a gas depends on its volume as kV^3. The work done when the temperature changes from 100°C to 300°C will be _____ nR, where n denotes number of moles of a gas.

Open solution

Q182. JEE Main 2020 (Online) 2nd September Morning Slot

An engine takes in 5 moles of air at 20^oC and 1 atm, and compresses it adiabatically to $1/10^{th}$ of the original volume. Assuming air to be a diatomic ideal gas made up of rigid molecules, the change in its internal energy during this process comes out to be X kJ. The value of X to the nearest integer is_____.

Open solution

Q183. JEE Main 2020 (Online) 9th January Evening Slot

Starting at temperature 300 K, one mole of an ideal diatomic gas (γ = 1.4) is first compressed adiabatically from volume V_1 to $V_2 = \frac{V_1}{16}$. It is then allowed to expand isobarically to volume $2V_2$. If all the processes are the quasi-static then the final temperature of the gas (in ^oK) is (to the nearest integer) ____.

Open solution

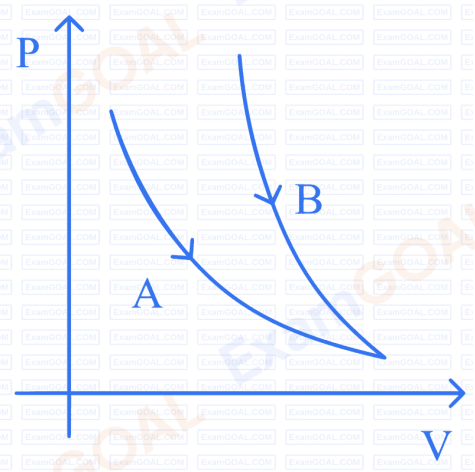
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MCQ (More than One Correct Answer) 2 questions

Q184. JEE Main 2024 (Online) 30th January Evening Shift

Open solution

Choose the correct statement for processes A & B shown in figure.



(A) $PV = k$ for process B and A.

(B) $\frac{P^{\gamma-1}}{T^{\gamma}} = k$ for process B and $T = k$ for process A.

(C) $\frac{T^{\gamma}}{P^{\gamma-1}} = k$ for process A and $PV = k$ for process B.

(D) $PV^{\gamma} = k$ for process B and $PV = k$ for process A.

Q185. AIEEE 2005

Open solution

Which of the following is **incorrect** regarding the first law of thermodynamics?

(A) It is a restatement of the principle of conservation of energy

(B) It is not applicable to any cyclic process

(C) It introduces the concept of the entropy

(D) It introduces the concept of the internal energy

Out of syllabus 46 questions

These questions are on topics no longer in the current syllabus. They are kept for practice, after everything else.

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Q186. JEE Main 2023 (Online) 12th April Morning Shift · MCQ (Single Correct Answer)

Open solution

An engine operating between the boiling and freezing points of water will have

A. efficiency more than 27%.

B. efficiency less than the efficiency of a Carnot engine operating between the same two temperatures.

C. efficiency equal to 27%

D. efficiency less than 27%

Choose the correct answer from the options given below:

(A) B and C only (B) B and D only (C) B, C and D only (D) A and B only

Q187. JEE Main 2023 (Online) 8th April Evening Shift · MCQ (Single Correct Answer)

Open solution

Work done by a Carnot engine operating between temperatures $127^{\circ}\mathrm{C}$ and $27^{\circ}\mathrm{C}$ is $2\,\mathrm{kJ}$. The amount of heat transferred to the engine by the reservoir is :

(A) 8 kJ (B) 2 kJ (C) 4 kJ (D) 2.67 kJ

Q188. JEE Main 2023 (Online) 1st February Evening Shift · MCQ (Single Correct Answer)

Open solution

A Carnot engine operating between two reservoirs has efficiency $\frac{1}{3}$. When the temperature of cold reservoir is raised by x, its efficiency decreases to $\frac{1}{6}$. The value of x, if the temperature of hot reservoir is 99°C, will be :

(A) 66 K (B) 62 K (C) 16.5 K (D) 33 K

Q189. JEE Main 2023 (Online) 30th January Evening Shift · MCQ (Single Correct Answer)

Open solution

Given below are two statements: one is labelled as Assertion $\mathbf{A}$ and the other is labelled as Reason $\mathbf{R}$

Assertion A: Efficiency of a reversible heat engine will be highest at $-273^{\circ}\mathrm{C}$ temperature of cold reservoir.

Reason R: The efficiency of Carnot's engine depends not only on the temperature of the cold reservoir but it depends on the temperature of the hot reservoir too and is given as $\eta = \left(1 - \frac{T_2}{T_1}\right)$

In the light of the above statements, choose the correct answer from the options given below

(A) Both $\mathbf{A}$ and $\mathbf{R}$ are true and $\mathbf{R}$ is the correct explanation of $\mathbf{A}$

(B) Both $\mathbf{A}$ and $\mathbf{R}$ are true but $\mathbf{R}$ is NOT the correct explanation of $\mathbf{A}$

(C) A is false but $\mathbf{R}$ is true

(D) A is true but $\mathbf{R}$ is false

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Q190. JEE Main 2023 (Online) 25th January Morning Shift · MCQ (Single Correct Answer)

A Carnot engine with efficiency 50% takes heat from a source at 600 K. In order to increase the efficiency to 70%, keeping the temperature of sink same, the new temperature of the source will be :

(A) 300 K (B) 1000 K (C) 900 K (D) 360 K

Open solution

Q191. JEE Main 2022 (Online) 30th July Evening Shift · MCQ (Single Correct Answer)

A Carnot engine is operating between the two reservoirs of temperature $227\ ^{\circ}\mathrm{C}$ and $127\ ^{\circ}\mathrm{C}$ respectively. If the engine performs $1.04 \times 10^5\ \mathrm{J}$ work per cycle. Then, the heat absorbed from hot reservoir per cycle will be :

(A) $2.08 \times 10^4\ \mathrm{J}$ (B) $1.04 \times 10^5\ \mathrm{J}$ (C) $5.20 \times 10^5\ \mathrm{J}$ (D) $2.36 \times 10^5\ \mathrm{J}$

Open solution

Q192. JEE Main 2022 (Online) 28th July Morning Shift · MCQ (Single Correct Answer)

A Carnot engine has efficiency of 50%. If the temperature of sink is reduced by $40^{\circ}\mathrm{C}$, its efficiency increases by 30%. The temperature of the source will be:

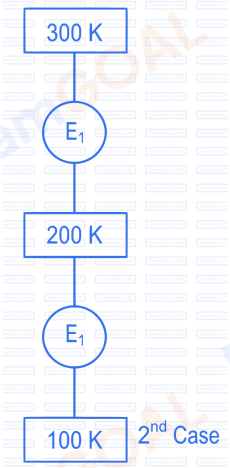
(A) 166.7 K (B) 255.1 K (C) 266.7 K (D) 367.7 K

Open solution

Q193. JEE Main 2022 (Online) 27th July Evening Shift · MCQ (Single Correct Answer)

In 1st case, Carnot engine operates between temperatures $300\ \mathrm{K}$ and $100\ \mathrm{K}$. In 2nd case, as shown in the figure, a combination of two engines is used. The efficiency of this combination (in 2nd case) will be :

Open solution



(A) same as the 1st case.

(B) always greater than the 1st case.

(C) always less than the 1st case.

(D) may increase or decrease with respect to the 1st case.

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Q194. JEE Main 2022 (Online) 25th July Evening Shift · MCQ (Single Correct Answer)

Let η_1 is the efficiency of an engine at $T_1 = 447^\circ C$ and $T_2 = 147^\circ C$ while η_2 is the efficiency at $T_1 = 947^\circ C$ and $T_2 = 47^\circ C$. The ratio $\frac{\eta_1}{\eta_2}$ will be :

(A) 0.41 (B) 0.56 (C) 0.73 (D) 0.70

Open solution

Q195. JEE Main 2022 (Online) 26th June Morning Shift · MCQ (Single Correct Answer)

The efficiency of a Carnot's engine, working between steam point and ice point, will be :

(A) 26.81% (B) 37.81% (C) 47.81% (D) 57.81%

Open solution

Q196. JEE Main 2022 (Online) 24th June Evening Shift · MCQ (Single Correct Answer)

A Carnot engine takes 5000 kcal of heat from a reservoir at 727°C and gives heat to a sink at 127°C. The work done by the engine is

(A) 3×10^6 J (B) Zero (C) 12.6×10^6 J (D) 8.4×10^6 J

Open solution

Q197. JEE Main 2022 (Online) 24th June Morning Shift · MCQ (Single Correct Answer)

A Carnot engine whose heat sink is at 27°C, has an efficiency of 25%. By how many degrees should the temperature of the source be changed to increase the efficiency by 100% of the original efficiency?

(A) Increases by 18°C (B) Increases by 200°C
(C) Increases by 120°C (D) Increases by 73°C

Open solution

Q198. JEE Main 2021 (Online) 31st August Morning Shift · MCQ (Single Correct Answer)

A reversible engine has an efficiency of $\frac{1}{4}$. If the temperature of the sink is reduced by 58°C, its efficiency becomes double. Calculate the temperature of the sink :

(A) 174°C (B) 280°C (C) 180.4°C (D) 382°C

Open solution

Q199. JEE Main 2021 (Online) 26th August Evening Shift · MCQ (Single Correct Answer)

A refrigerator consumes an average 35W power to operate between temperature $-10^\circ C$ to $25^\circ C$. If there is no loss of energy then how much average heat per second does it transfer?

(A) 263 J/s (B) 298 J/s (C) 350 J/s (D) 35 J/s

Open solution

Q200. JEE Main 2021 (Online) 27th July Evening Shift · MCQ (Single Correct Answer)

Two Carnot engines A and B operate in series such that engine A absorbs heat at T_1 and rejects heat to a sink at temperature T. Engine B absorbs half of the heat rejected by Engine A and rejects heat to the sink at T_3. When workdone in both the cases is equal, to value of T is :

(A) $\frac{2}{3}T_1 + \frac{3}{2}T_3$ (B) $\frac{1}{3}T_1 + \frac{2}{3}T_3$
(C) $\frac{3}{2}T_1 + \frac{1}{3}T_3$ (D) $\frac{2}{3}T_1 + \frac{1}{3}T_3$

Open solution

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Q201. JEE Main 2021 (Online) 25th July Evening Shift · MCQ (Single Correct Answer)

A heat engine has an efficiency of $\frac{1}{6}$. When the temperature of sink is reduced by 62°C, its efficiency gets doubled. The temperature of the source is :

(A) 124°C (B) 37°C (C) 62°C (D) 99°C

Open solution

Q202. JEE Main 2021 (Online) 17th March Morning Shift · MCQ (Single Correct Answer)

A Carnot's engine working between 400 K and 800 K has a work output of 1200 J per cycle. The amount of heat energy supplied to the engine from the source in each cycle is :

(A) 2400 J (B) 1600 J (C) 1800 J (D) 3200 J

Open solution

Q203. JEE Main 2020 (Online) 2nd September Evening Slot · MCQ (Single Correct Answer)

A heat engine is involved with exchange of heat of 1915 J, − 40J, + 125 J and −Q J, during one cycle achieving an efficiency of 50.0%. The value of Q is

(A) 980 J (B) 40 J (C) 400 J (D) 640 J

Open solution

Q204. JEE Main 2020 (Online) 8th January Evening Slot · MCQ (Single Correct Answer)

A carnot engine having an efficiency of $\frac{1}{10}$ is being used as a refrigerator. If the work done on the refrigerator is 10 J, the amount of heat absorbed from the reservoir at lower temperature is :

(A) 90 J (B) 99 J (C) 1 J (D) 100 J

Open solution

Q205. JEE Main 2020 (Online) 7th January Evening Slot · MCQ (Single Correct Answer)

Two ideal Carnot engines operate in cascade (all heat given up by one engine is used by the other engine to produce work) between temperature, T_1 and T_2 . The temperature of the hot reservoir of the first engine is T_1 and the temperature of the cold reservoir of the second engine is T_2 . T is temperature of the sink of first engine which is also the source for the second engine. How is T related to T_1 and T_2 , if both engines perform equal amount of work?

(A) $T = \frac{2T_1T_2}{T_1+T_2}$ (B) $T = \sqrt{T_1T_2}$

(C) $T = \frac{T_1+T_2}{2}$ (D) T = 0

Open solution

Q206. JEE Main 2019 (Online) 12th April Evening Slot · MCQ (Single Correct Answer)

A Carnot engine has an efficiency of $\frac{1}{6}$. When the temperature of the sink is reduced by 62°C, its efficiency is doubled. The temperatures of the source and the sink are, respectively

(A) 99°C, 37°C (B) 124°C, 62°C (C) 37°C, 99°C (D) 62°C, 124°C

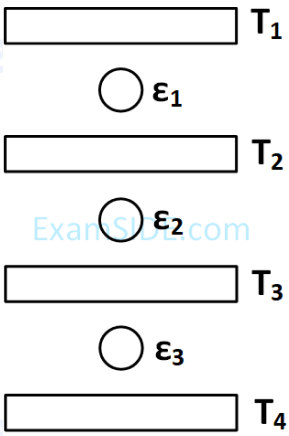
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Q207. JEE Main 2019 (Online) 10th January Morning Slot · MCQ (Single Correct Answer)

Three Carnot engines operate in series between a heat source at a temperature T_1 and a heat sink at temperature T_4 (see figure). There are two other reservoirs at temperature T_2 and T_3, as shown, with $T_1 > T_2 > T_3 > T_4$. The three engines are equally efficient if -

Open solution



(A) $T_2 = (T_1^3T_4)^{1/4}$; $T_3 = (T_1T_4^3)^{1/4}$

(B) $T_2 = (T_1T_4)^{1/2}$; $T_3 = (T_1^2T_4)^{1/3}$

(C) $T_2 = (T_1T_4^2)^{1/3}$; $T_3 = (T_1^2T_4)^{1/3}$

(D) $T_2 = (T_1^2T_4)^{1/3}$; $T_3 = (T_1T_4^2)^{1/3}$

Q208. JEE Main 2019 (Online) 9th January Evening Slot · MCQ (Single Correct Answer)

Two Carnot engines A and B are operated in series. The first one, A, receives heat at T_1 (= 600 K) and rejects to a reservoir at temperature T_2. The second engine B receives heat rejected by the first engine and, in turn, rejects to a heat reservoir at T_3 (=400 K). Calculate the temperature T_2 if the work outputs of the two engines are equal :

Open solution

(A) 600 K

(B) 400 K

(C) 300 K

(D) 500 K

Q209. JEE Main 2018 (Online) 15th April Evening Slot · MCQ (Single Correct Answer)

Two Carnot engines A and B are operated in series. Engine A receives heat from a reservoir at 600 K and rejects heat to a reservoir at temperature T. Engine B receives heat rejected by engine A and in turn rejects it to a reservoir at 100 K. If the efficiencies of the two engines A and B are represented by η_A and η_B, respectively, then what is the value of $\frac{\eta_B}{\eta_A}$?

Open solution

(A) $\frac{12}{7}$

(B) $\frac{7}{12}$

(C) $\frac{12}{5}$

(D) $\frac{5}{12}$

Q210. JEE Main 2018 (Online) 15th April Morning Slot · MCQ (Single Correct Answer)

A Carnot's engine works as a refrigerator between 250 K and 300 K. It receives 500 cal heat from the reservoir at the lower temperature. The amount of work done in each cycle to operate the refrigerator is :

Open solution

(A) $420\ J$

(B) $772\ J$

(C) $2100\ J$

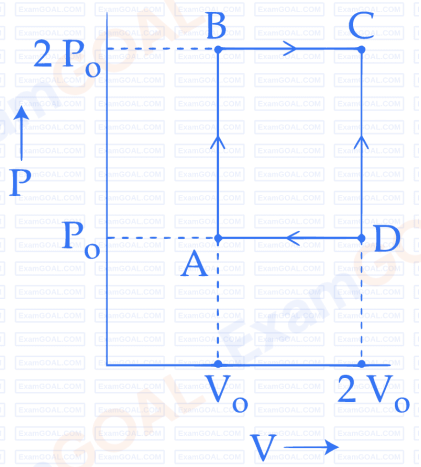
(D) $2520\ J$

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Q211. JEE Main 2017 (Online) 8th April Morning Slot · MCQ (Single Correct Answer)

An engine operates by taking n moles of an ideal gas through the cycle ABCDA shown in figure. The thermal efficiency of the engine is : (Take C_V = 1.5 R, where R is gas constant)

Open solution



(A) 0.24 (B) 0.15 (C) 0.32 (D) 0.08

Q212. JEE Main 2016 (Online) 10th April Morning Slot · MCQ (Single Correct Answer)

A Carnot freezer takes heat from water at 0^oC inside it and rejects it to the room at a temperature of 27^oC. The latent heat of ice is 336×10^3 J kg^{-1}. If 5 kg of water at 0^oC is converted into ice at 0^oC by the freezer, then the energy consumed by the freezer is close to :

Open solution

(A) 1.67×10^5 J (B) 1.68×10^6 J (C) 1.51×10^5 J (D) 1.71×10^7 J

Q213. JEE Main 2014 (Online) 12th April Morning Slot · MCQ (Single Correct Answer)

A Carnot engine absorbs 1000 J of heat energy from a reservoir at 127^oC and rejects 600 J of heat energy during each cycle. The efficiency of engine and temperature of sink will be :

Open solution

(A) 20% and -43^oC (B) 40% and -33^oC (C) 50% and -20^oC (D) 70% and -10^oC

Q214. AIEEE 2012 (Online) 26th May · MCQ (Single Correct Answer)

The door of a working refrigerator is left open in a well insulated room. The temperature of air in the room will

Open solution

(A) decrease

(B) increase in winters and decrease in summers

(C) remain the same

(D) increase

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Q215. AIEEE 2012 (Online) 19th May · MCQ (Single Correct Answer)

Open solution

This question has Statement 1 and Statement 2. Of the four choices given after the Statements, choose the one that best describes the two Statements.
Statement 1: An inventor claims to have constructed an engine that has an efficiency of 30% when operated between the boiling and freezing points of water. This is not possible.
Statement 2: The efficiency of a real engine is always less than the efficiency of a Carnot engine operating between the same two temperatures.

(A) Statement 1 is true, Statement 2 is true, Statement 2 is not the correct explanation of Statement 1.

(B) Statement 1 is true, Statement 2 is false.

(C) Statement 1 is false, Statement 2 is true.

(D) Statement 1 is true, Statement 2 is true, Statement 2 is the correct explanation of Statement 1.

Q216. AIEEE 2012 · MCQ (Single Correct Answer)

Open solution

A Carnot engine, whose efficiency is 40%, takes in heat from a source maintained at a temperature of $500\ K$. It is desired to have an engine of efficiency 60%. Then, the intake temperature for the same exhaust (sink) temperature must be :

(A) efficiency of Carnot engine cannot be made larger than 50%

(B) $1200\ K$

(C) $750\ K$

(D) $600\ K$

Q217. AIEEE 2011 · MCQ (Single Correct Answer)

Open solution

A Carnot engine operating between temperatures T_1 and T_2 has efficiency $\frac{1}{6}$. When T_2 is lowered by $62\ K$ its efficiency increases to $\frac{1}{3}$. Then T_1 and T_2 are, respectively:

(A) $372\ K$ and $330\ K$ (B) $330\ K$ and $268\ K$ (C) $310\ K$ and $248\ K$ (D) $372\ K$ and $310\ K$

Q218. AIEEE 2010 · MCQ (Single Correct Answer)

Open solution

A diatomic ideal gas is used in a Carnot engine as the working substance. If during the adiabatic expansion part of the cycle the volume of the gas increases from V to $32\ V$, the efficiency of the engine is

(A) 0.5 (B) 0.75 (C) 0.99 (D) 0.25

Q219. AIEEE 2007 · MCQ (Single Correct Answer)

Open solution

A Carnot engine, having an efficiency of $\eta = 1/10$ as heat engine, is used as a refrigerator. If the work done on the system is $10\ J$, the amount of energy absorbed from the reservoir at lower temperature is

(A) $100\ J$ (B) $99\ J$ (C) $90\ J$ (D) $1\ J$

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Q220. AIEEE 2003 · MCQ (Single Correct Answer)

A carnot engine takes 3×10^6 cal. of heat from a reservoir at $627^\circ C$, and gives it to a sink at $27^\circ C$. The work done by the engine is

(A) $4.2 \times 10^6 J$ (B) $8.4 \times 10^6 J$ (C) $16.8 \times 10^6 J$ (D) zero

Open solution

Q221. AIEEE 2002 · MCQ (Single Correct Answer)

Which statement is incorrect?

(A) all reversible cycles have same efficiency

(B) reversible cycle has more efficiency than an irreversible one

(C) Carnot cycle is a reversible one

(D) Carnot cycle has the maximum efficiency in all cycles.

Open solution

Q222. AIEEE 2002 · MCQ (Single Correct Answer)

Even Carnot engine cannot give 100% efficiency because we cannot

(A) prevent radiation (B) find ideal sources

(C) reach absolute zero temperature (D) eliminate friction.

Open solution

Q223. JEE Main 2022 (Online) 29th June Morning Shift · Numerical

300 cal. of heat is given to a heat engine and it rejects 225 cal. If source temperature is 227°C, then the temperature of sink will be _____ °C.

Open solution

Q224. JEE Main 2022 (Online) 27th June Morning Shift · Numerical

In a carnot engine, the temperature of reservoir is 527°C and that of sink is 200 K. If the work done by the engine when it transfers heat from reservoir to sink is 12000 kJ, the quantity of heat absorbed by the engine from reservoir is _____ $\times 10^6$ J.

Open solution

Q225. JEE Main 2022 (Online) 26th June Evening Shift · Numerical

A heat engine operates with the cold reservoir at temperature 324 K. The minimum temperature of the hot reservoir, if the heat engine takes 300 J heat from the hot reservoir and delivers 180 J heat to the cold reservoir per cycle, is _____ K.

Open solution

Q226. JEE Main 2021 (Online) 27th August Evening Shift · Numerical

A heat engine operates between a cold reservoir at temperature T_2 = 400 K and a hot reservoir at temperature T_1. It takes 300 J of heat from the hot reservoir and delivers 240 J of heat to the cold reservoir in a cycle. The minimum temperature of the hot reservoir has to be _____ K.

Open solution

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Q227. JEE Main 2021 (Online) 16th March Evening Shift · **Numerical**

For an ideal heat engine, the temperature of the source is 127°C. In order to have 60% efficiency the temperature of the sink should be _____°C. (Round off to the Nearest Integer)

Open solution

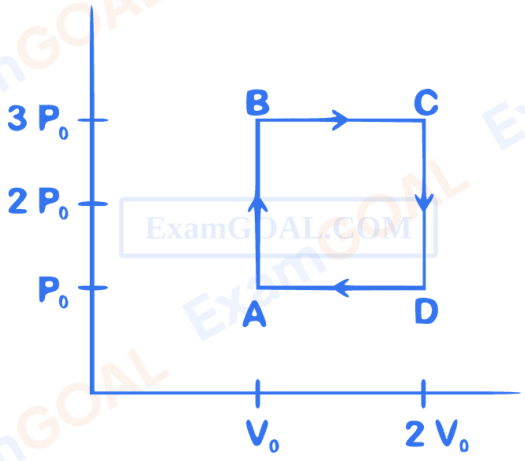
Q228. JEE Main 2021 (Online) 25th February Evening Shift · **Numerical**

A reversible heat engine converts one-fourth of the heat input into work. When the temperature of the sink is reduced by 52K, its efficiency is doubled. The temperature in Kelvin of the source will be _____.

Open solution

Q229. JEE Main 2020 (Online) 6th September Evening Slot · **Numerical**

An engine operates by taking a monoatomic ideal gas through the cycle shown in the figure. The percentage efficiency of the engine is close to _____.



Open solution

Q230. JEE Main 2020 (Online) 3rd September Evening Slot · **Numerical**

If minimum possible work is done by a refrigerator in converting 100 grams of water at 0°C to ice, how much heat (in calories) is released to the surroundings at temperature 27°C (Latent heat of ice = 80 Cal/gram) to the nearest integer ?

Open solution

Q231. JEE Main 2020 (Online) 7th January Morning Slot · **Numerical**

A Carnot engine operates between two reservoirs of temperatures 900 K and 300 K. The engine performs 1200 J of work per cycle. The heat energy (in J) delivered by the engine to the low temperature reservoir, in a cycle, is _____.

Open solution

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Answer Key

MCQ (Single Correct Answer)

1. **B**	2. **C**	3. **B**	4. **C**	5. **C**
6. **B**	7. **D**	8. **D**	9. **A**	10. **A**
11. **A**	12. **C**	13. **D**	14. **B**	15. **C**
16. **B**	17. **B**	18. **A**	19. **C**	20. **B**
21. **B**	22. **B**	23. **D**	24. **B**	25. **B**
26. **D**	27. **C**	28. **C**	29. **B**	30. **D**
31. **B**	32. **D**	33. **B**	34. **B**	35. **B**
36. **B**	37. **C**	38. **C**	39. **A**	40. **C**
41. **B**	42. **A**	43. **B**	44. **C**	45. **D**
46. **B**	47. **A**	48. **A**	49. **B**	50. **C**
51. **C**	52. **D**	53. **D**	54. **A**	55. **C**
56. **C**	57. **B**	58. **D**	59. **D**	60. **C**
61. **B**	62. **B**	63. **D**	64. **B**	65. **C**
66. **B**	67. **B**	68. **B**	69. **C**	70. **B**
71. **C**	72. **B**	73. **B**	74. **D**	75. **C**
76. **B**	77. **A**	78. **B**	79. **A**	80. **A**
81. **B**	82. **B**	83. **D**	84. **B**	85. **A**
86. **C**	87. **D**	88. **A**	89. **B**	90. **A**
91. **A**	92. **C**	93. **B**	94. **1**	95. **C**
96. **A**	97. **B**	98. **B**	99. **A**	100. **B**
101. **A**	102. **C**	103. **A**	104. **B**	105. **C**
106. **A**	107. **C**	108. **D**	109. **C**	110. **A**
111. **B**	112. **B**	113. **B**	114. **D**	115. **C**
116. **A**	117. **B**	118. **D**	119. **C**	120. **C**
121. **D**	122. **D**	123. **D**	124. **C**	125. **A**
126. **D**	127. **A**	128. **A**	129. **C**	130. **A**
131. **D**	132. **B**	133. **C**	134. **C**	135. **B**
136. **C**	137. **B**	138. **B**	139. **D**	140. **D**
141. **C**	142. **D**	143. **B**	144. **A**	145. **C**
146. **B**	147. **A**	148. **A**	149. **A**	150. **B**

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151. **B**	152. **A**	153. **B**	154. **D**	155. **B**
156. **A**	157. **A**	158. **D**	159. **C**	

Numerical

160. **300**	161. **120**	162. **300**	163. **481**	164. **350**
165. **500**	166. **314**	167. **15**	168. **273**	169. **4**
170. **750**	171. **1400**	172. **2**	173. **12**	174. **25**
175. **480**	176. **17258**	177. **100**	178. **25**	179. **60**
180. **3600**	181. **50**	182. **46**	183. **1818 to 1819**	

MCQ (More than One Correct Answer)

184. **B, D**	185. **B, C**

Out of syllabus

186. **B**	187. **A**	188. **B**	189. **A**	190. **B**
191. **C**	192. **C**	193. **A**	194. **B**	195. **A**
196. **C**	197. **B**	198. **A**	199. **A**	200. **D**
201. **D**	202. **A**	203. **A**	204. **A**	205. **C**
206. **A**	207. **D**	208. **D**	209. **A**	210. **A**
211. **B**	212. **A**	213. **B**	214. **D**	215. **D**
216. **C**	217. **D**	218. **B**	219. **C**	220. **B**
221. **A**	222. **C**	223. **102**	224. **16**	225. **540**
226. **500**	227. **-113**	228. **208**	229. **19**	230. **8791**
231. **600**				

For numerical answers given as a range, any value within the range is correct. Full solutions: scan the QR code next to each question, or visit questions.examside.com.

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